

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

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Role. Senior Supply Chain Manager, NACF Supply Chain — Global Capacity Planning team. Job ID 3195330. Bellevue, WA.

Team mission, in their words. Plan and deliver North America Customer Fulfillment capacity to meet customer demand at the desired speed and quality while improving the long-term cost structure of the network.

Their customers. S&OP, PPT, Finance, FC Operations, WW and Operations Engineering, Real Estate, WW Design, Launch, Safety/EHS. Partners in SCOT, Product, Engineering and Construction.

Format. One hour. Expect two to four Leadership Principle stories with hard follow-ups, and one functional block on capacity planning. If this is the hiring manager, the functional block gets longer.

Your position. You have designed and consolidated DC capacity, allocated demand across a 3-DC / 100-store network through peak, and turned measured lane data into planning parameters. You have not built an Amazon FC capacity plan. Say so once, early, then show the mechanism.

Read sections 2, 3, and 4 out loud once — section 3 has "tell me about yourself" in the Amazon shape and "why Amazon / why this team / why now." Section 5 is the story bank in STAR with the interviewer's own probes answered — know which four you are leading with. Section 6 has the question-bank index: hear the question, know the story. Section 7A is the deep model — how a unit flows, what it costs, and the WBR this role would run; read it once, slowly. Section 7 is the functional block; walk Case A on paper once before Friday.

1. What they are actually scoring

The L6 bar, in one table

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Ambiguity. Strategy may not be defined. You pick the stakeholders, set the goals, design the long-term solution.	Did you start from a vague ask and define the problem yourself?	Elementum: the ask was a report; you built lane-level lead-time planning. Flexport: inherited an 18-month plan and re-costed the sequence.
Scope and influence. A team, sometimes an org or country. Influences 1–2 levels up.	Did you move a decision above your level with evidence, not seniority?	Roadmap flip ratified in Bellevue after private math with your manager. CIO/COO weekly pack across 60+ stakeholders.
Execution. Tactical and strategic. Decides where to simplify or extend. Mitigates long-term risk. Trade-offs short vs long.	Can you name the trade-off and the number that decided it?	Partner-first vs operator-first (CTS ~50% vs ~25%). Robots at 30% vs the launch date. Mock tolerance vs go-live.
Complex. Visible risks, roadblocks, constraints. You design an appropriately simple solution.	Was the solution simpler than the problem? Did a gate exist before the crisis?	0.5% tolerance published before mock one. 98% pick-accuracy gate published before the robots arrived.
Designing or modeling the supply chain, not working in it.	Are your stories about the network's shape and parameters, or about running a shift?	Allocation across 3 DCs + 100 stores. Two buildings into one omni DC. 200 lanes measured. Safety-stock parameters re-set from data.

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Process improvement. Drives operational-excellence best practice.	Did you leave a mechanism that ran without you?	WBR grammar at Flexport; QA-authored release gate that survived your leaving; carrier escalation and drift flags.

How the interviewer is trained to probe — Amazon's STAR worksheet

Amazon's interviewer guide gives every interviewer the same worksheet: establish the story in STAR, then **probe** on the assigned Leadership Principle, then **challenge** the candidate's decisions. Results are scored against a legend — \$ cost savings or revenue, # volume/scale, % change, **time** saved, **customer/team impact**, **quality**. Every story in Section 5 is written so the probes below are already answered before they are asked, and every Result is written to that legend.

PHASE	WHAT THEY ARE TRAINED TO ASK	HOW YOUR ANSWER PRE-EMPTS IT
Situation / Task	Why is this important? What was the goal? What was the initial scope? What were the challenges? What were the risks if nothing happened?	Open every story with <i>problem</i> → <i>goal</i> → <i>the metric and its baseline</i> → <i>the cost of doing nothing</i> . Fifteen seconds. Then stop describing and start doing.
Action	Were you the key driver? Biggest contribution? Unique value? Most significant obstacles? What did you do versus the team? How did you set priorities / get buy-in? What decisions did you challenge?	Every action sentence starts with "I" and names an artifact — the gate, the table, the rule, the log, the query. Name the obstacle and the decision you overruled. Name what you deliberately did <i>not</i> do.
Results	How did you measure success? Why these results? What percentage change year over year? What trade-offs (quality, cost, time)? "I'm concerned about... tell me more."	Give the metric, grain, baseline, target, actual — in that order. Then the trade-off you paid. Then the mechanism that made it stick.
Challenge	Why this story? Can you give a more recent one? What were the lessons? What would you do differently? How would you implement this at Amazon? How did results compare to the goal you stated?	Every story below ends with "Did the metric move" against the stated goal, a real "differently," and an "At Amazon" sentence that lands in capacity planning.

Three words they must hear in every story: supply, demand, capacity. Each story in Section 5 carries a box naming what supply was, what demand was, what capacity was, and which one bound. Say the box out loud when the question is functional; keep it in your head when the question is behavioral — but make sure the story's logic follows it.

The JD, translated into what you say

- **"Take end-to-end ownership of building and manifesting our capacity plan."** Say: a plan is a set of numbers with named owners and triggers, not a deck. You have run that as a WBR.
- **"Balance capacity and demand."** Say: that is allocation at Manhattan and consolidation at Sephora UDC. Demand did not fit the building; you changed where units went, then changed the building.
- **"Define metrics, develop program management mechanisms, invent/simplify processes."** Say: input lever vs output metric, pre-wired reds, COE that changes a mechanism. Flexport Snowflake story.
- **"Craft concise written narratives to influence executive decisions."** Say: the two-column cost-to-serve table that flipped an 18-month roadmap. Offer to write it.
- **"Adherence to lean philosophy at scale."** Say: gates that can stop the line. Andon is not a poster.

Vocabulary you should use correctly, and what to ask about

TERM	USE IT AS
NACF	North America Customer Fulfillment — the FC network that picks, packs and ships customer orders.
Capacity plan	Units of demand the network must absorb, by week, by building type, against what each building can store and push out. Two constraints, not one: storage (cube) and throughput (units per hour through inbound, stow, pick, pack, ship). Say both every time.
S&OP	The weekly plan that reconciles demand forecast, inventory and capacity; the basis of labor planning and steering decisions. Your Elementum forecast-alert story lives here.
OP1 / OP2	Amazon's annual operating-plan cycles (fall and spring). Long-range capacity decisions — new builds, expansions, retrofits — are argued in these. You can say "the OP cycle" without pretending you have sat in one.
SCOT	Supply Chain Optimization Technologies — the science and product org that owns forecasting, buying, placement and planning systems. The interface you would manage: they own the model, you own whether the plan is honest.
PPT	Listed as a customer of the team. Do not guess the acronym. Ask: "what does PPT own relative to Global Capacity Planning?"
Sortable / non-sortable / AR	FC types by item size and automation. Capacity is planned per type; a sortable AR building and a non-sort building are not interchangeable. Say "by building type" rather than "by FC."
Launch / Real Estate / WW Design	The teams that put new capacity on the ground. Your UDC and dual-DC stories are launch stories; speak to them in launch gates and freeze windows.
Speed	Promised delivery speed to the customer. Capacity in the wrong place is a speed miss, not just a cost miss. Your promise-equation work maps straight onto this.

2. Say this about the gap. Once.

UNPROMPTED, INSIDE THE FIRST FIVE MINUTES

"I have not built an Amazon FC capacity plan. I have done the three things underneath one: I allocated demand across three mega-DCs and a hundred stores through peak when the network did not fit the plan; I consolidated two warehouses into one omni facility with robotics and ran the launch gates; and I replaced assumed planning parameters with measured ones across two hundred lanes and released the inventory the bad assumptions were holding. I know the scale here is different. The mechanism is not."

Then stop. Do not list what else you have not done. If they probe the gap later, the answer is in section 8 ("Why you over an internal transfer").

3. "Tell me about yourself" — Amazon format, 60 seconds

Present → past → future. One sentence each for present and future; three numbered chapters for past, one number each. Under 160 words. No adjectives about yourself. Say the three words — **supply, demand, capacity** — once, in the last line, where they land as a claim about the job.

SAY THIS, THEN STOP — ABOUT 60 SECONDS

Present. "I'm the technical program manager for retail fulfillment and inventory at Sephora — I own how inventory is defined, placed and promised across stores, DC and dotcom on S/4HANA, and I just consolidated two warehouses

into one robotic omni DC: processing efficiency up fifteen percent, zero customer incidents through peak.

Past. Before that, three chapters, one job. Manhattan Associates: allocation and promising across three mega-DCs and a hundred stores — 99% fill through Cyber Monday, about eight million protected. Elementum: replaced brochure lead times with measured distributions on two hundred lanes — a million and a half of safety stock released at 99% accuracy. Flexport: flipped an eighteen-month roadmap with a two-column cost table — five million avoided, value in six months.

Future. I have not built an Amazon FC capacity plan. What I have done four times is plan supply, demand and capacity as one equation, deliver it through gates, and keep it honest in a weekly review. I want to do that at network scale, in the one place where it is the operating system."

If they want the long version, add one sentence per chapter from the table in Section 11. Do not add adjectives; add numbers. **If they interrupt**, the last two sentences are the only ones that must land.

Why Amazon. Why this team. Why now.

The generic version — "customer obsession, scale, Leadership Principles" — scores zero because every candidate says it. The specific version names one thing Amazon did, one thing you did, and the exact seam where they meet. Under 45 seconds each. Items marked **[public]** are from Amazon's public statements; re-read the source before Friday and do not quote a number you have not re-read.

WHY AMAZON — SPECIFIC

"Three specific reasons, none of them 'because it's Amazon.' First: regionalization **[public — 2023 shareholder letter]**. When Amazon reorganized the U.S. fulfillment network into regions so that most orders ship from inside the customer's own region, it got faster *and* cheaper at the same time — fewer touches, shorter distance. I read that as a capacity-planning decision, not a transportation one: where you put cube and hours decided both the speed line and the cost line. That is the exact trade I made at one-thousandth the scale when I consolidated two Sephora warehouses, and when I re-costed Flexport's roadmap on cost-to-serve. I want to make it at the scale where it is the company's P&L. Second: the review. For six years I have run weekly reviews in borrowed Amazon grammar — input metrics, output metrics, reds pre-decomposed, a correction of error that changes a mechanism. The Flexport roadmap decision I am proudest of was argued on a two-column cost table in a Bellevue onsite and on nothing else. Here that is the default, not something I have to install. Third: the demand signal. Everywhere I have planned capacity, the forecast was a spreadsheet I had to defend. Here it is a model with an error distribution. That changes what a capacity planner can promise — you can plan to a quantile instead of to an argument."

WHY THIS TEAM — GLOBAL CAPACITY PLANNING

"The charter has both halves of the only trade I care about: meet demand at the promised speed, *and* improve the long-term cost structure. Most roles own one half and lobby against the other; this one owns the dial. And the customer list — S&OP, Finance, FC Ops, Engineering, Real Estate, Launch — is the set of rooms where I have already had the hard version of the conversation: handed S&OP a forecast alert they trusted enough to act on; given Finance a cost table and let them attack it; held a launch gate against a go-live date. I know how each room argues. I want the seat that has to get them to one number — and the first thing I would want to understand is where the storage-versus-throughput call actually gets made today, because that is where the plan is either honest or not."

WHY NOW — AND WHY LEAVE SEPHORA

"I've delivered what I went to Sephora to do: the WMS migration is live, the omni DC is live and fifteen percent more efficient, the inventory operating model is the standing reference. If I stay I run the same play a fourth time at retail scale. I would rather run it once at this one, where supply, demand and capacity are planned for a network of hundreds

of buildings and the forecast is a model. I'm not running from anything. I'm running toward the version of this problem where the network is the product."

Do not say: "most customer-centric company," "I love the Leadership Principles," "Day 1," or anything about compensation or Bellevue here. **If asked "why not stay in retail":** the last three sentences of "Why now."

4. The sixty-minute clock

MINUTE	WHAT IS HAPPENING	YOUR JOB
0–3	Intros. They describe the team.	Listen for the word they use for their pain: "peak," "utilization," "launch," "forecast," "cost per unit." That word picks your lead story.
3–7	"Tell me about yourself." Often followed by "why Amazon."	Section 3, Amazon format. Land the gap sentence. Keep "why Amazon" under 45 seconds.
6–40	Two to four LP questions. Each gets four or five follow-ups.	One story per question, 90 seconds, then stop and let them drill. Answer follow-ups in one or two sentences. Say "I" for decisions. Name the data pull.
40–52	Functional block: build a plan, handle a gap, define metrics.	Sections 7A and 7. Clarify out loud for 60 seconds before you solve. Say both currencies — cube and hours. Do a math checkpoint. End with owners and triggers.
52–58	Your questions.	Two from section 9. Not four.
58–60	Close.	Section 10.

Amazon interviewers write notes the whole time and score after. Pauses are fine. A tight 90 seconds followed by "happy to go deeper on any part" beats a four-minute story every time.

5. The story bank — STAR, probed to the worksheet

Sixteen stories. Every one has the same skeleton, in this order, because that is the order the interviewer's worksheet scores: **Problem** → **Goal** → **Key metric (baseline → target)** → **Situation** → **Task** → **Action** → **Result** → **Did the metric move?** Then the box naming **supply, demand and capacity** in that story. Then "what was innovative" — the usual approach versus what you did — because "actions" are scored on whether a competent peer would have done the same thing. Then the probes, grouped exactly as the worksheet groups them: Situation, Action, Results, Challenge. Then one sentence on how you would implement it at Amazon, because the worksheet tells them to ask.

Every number is from your resume or your war book. Where a probe answer describes *how* you computed a number, check the method against your memory before Friday — change the method if needed, keep the number. "AIQB" references are the question numbers in Amazon's interviewer question bank, so you can hear a question and know which story it was written for. The spoken answer is 90–120 seconds. Then stop and let them probe.

Lead with these four unless their pain word says otherwise: S1 (consolidation), S2 (peak allocation), S3 (200 lanes), S7 (roadmap flip). If their pain word is "inventory," "placement," "returns" or "cutover," S10 moves into the first four.

S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived (Sephora UDC)

LPs: Insist on the Highest Standards · Customer Obsession · Bias for Action · Deliver Results. **AIQB:** Highest Standards 1, 4, 6 · Deliver Results 1, 2 · Bias for Action 1 · Customer Obsession 8.

Problem	Store replenishment and dotcom ran out of two warehouses: the same SKU stowed twice, two labor pools, two inbound docks for one vendor base, and no shared view of what was sellable to whom. Cost per unit above plan and a transfer every time one channel needed the other's stock.
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Goal	One omnichannel DC on Blue Yonder with Locus robotics, live before peak, with zero customer-facing incidents and pick accuracy ≥ 98% before the robots carried full volume.
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Key metric	Processing efficiency (units per labor hour) vs the two-building baseline; pick accuracy by zone by hour vs 98%; customer-facing incidents.
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Did it move?	Yes. Processing efficiency +15%. Pick accuracy 94% → ≥ 98% within one week of the throttle. Zero customer-facing incidents through peak. Launched on date.
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Situation. "Sephora served stores and dotcom from two separate warehouses. Same SKUs stowed twice, two labor pools, one vendor base shipping to two docks, and no single answer to 'how many can we sell.' The decision was to collapse both into one omni DC on Blue Yonder with Locus robots — inside a window that ended at peak.

Task. I owned the program: fifteen-plus tracks, sixty-plus stakeholders, about a dozen systems — WMS, ESB and Kafka, OMS, finance, the data platform. I set the goal in numbers before we started: live before peak, zero customer-facing incidents, pick accuracy at or above 98% by zone before robots took full volume, processing efficiency above the two-building baseline.

Action. Three things that were not in the standard playbook. First, I published the pass/fail numbers before the robotics vendor was on site — 98% pick accuracy measured by zone by hour, not a building average — so the go/no-go would be read, not argued. Second, I designed the robot ramp as a dial, not a switch: the manual pick path stayed fully staffed and certified through the ramp, volume could move between bots and people within a shift, and the decision right for that dial sat with me. Third, one issues log — 836 rows over fourteen months, every row an owner and a date — behind dated public test gates, November 22, December 20, January 30, and an October-to-December

change freeze so the building would survive peak. Week two after robots went live, pick accuracy printed 94%. I throttled robots to about 30% of volume the same day — only the zones that had cleared 98% in the debug sample — kept the launch green on the manual path, sat the zone-logic debug with the vendor, and restored full volume in week three when every zone cleared the bar.

Result. Processing efficiency finished plus fifteen percent against the two-building baseline. Pick accuracy from 94% to above 98% in a week. Zero customer-facing incidents through peak. Launched on the date. And structurally: one inventory, one dock, one labor pool — a unit can serve a store or a customer order without a transfer. The gate existed so the throttle needed no meeting."

Supply · Demand · Capacity. **Demand:** two profiles — store replenishment (lumpy, planned) and dotcom (daily, peaky) — now served from one pool. **Supply:** one vendor inbound stream to one dock instead of two. **Capacity:** two buildings' cube and hours collapsed into one, with robotics as the throughput multiplier. **What bound at launch:** pick throughput *at quality*, not cube — the building had the space; the question was whether the bots could pick at 98%. The throttle was a throughput-capacity decision made against a quality gate.

What was innovative. The usual approach is a cutover date plus a readiness checklist and a vendor who says "ready." I made the quality bar a number published before the technology existed in the building, built the ramp as a reversible dial with named decision rights, and kept a certified manual path as paid insurance. That turned a potential launch crisis into a routine Tuesday decision.

SITUATION PROBES

Why was this important? What was the goal? Every unit stowed twice is cube paid for twice and a transfer waiting to happen. The business case was lower cost per unit and one view of sellable inventory; my program goal was the three numbers — zero incidents, $\geq 98\%$ by zone, efficiency above baseline — because a consolidation that ships wrong units has destroyed the reason it was built.

What would have happened if nothing was done? Two buildings through another peak with inter-DC transfers to cover channel imbalances, duplicated safety stock, and a cost per unit that stayed above plan. The lease and labor math did not work for two buildings at our volume.

What was the initial scope? Did it change? WMS, robotics, and inventory merge. Scope grew to the integration layer when we found store and dotcom order flows hit the WMS through different interfaces with different unit-of-measure conventions. I added the track and re-baselined the gate dates once, publicly, in writing.

Biggest challenge? Time. The window ended at peak and peak was non-negotiable. Everything was designed to be safe to stop: the manual path, the freeze, the gates.

ACTION PROBES

What did you personally do versus the team? I wrote the gate definition and published it; I made the throttle call the day 94% printed; I owned the issues log and the change freeze; I decided which zones the bots kept. The vendor fixed the zone logic; ops ran the manual path; I decided when volume moved.

Why 30%? How did you arrive at it? It was an output, not a target. I let the bots keep only the zones that had already cleared 98% in the debug sample. Those zones carried about 30% of volume. Anything else would have been a guess dressed as a plan.

Why not delay the robotics go-live until the vendor was confident? Vendor confidence is not a metric. I had a measured path to run both and a number that said when to move volume. Delaying would have spent calendar we did not have and taught us nothing about the zone logic under real volume.

Why not push through — the business had bought robots? A gate that cannot stop the program is decoration. The robots were an input; the customer-facing output was accuracy. I will not ship a customer-facing miss to protect an automation narrative.

What did the manual path cost? Who paid? About three weeks of labor hours for a certified path we were planning to shrink. I priced it against one week of accuracy misses at peak — mis-ships, re-ships, concessions,

store out-of-stocks — and against pulling the launch. It was the cheapest insurance in the program and I said so in the steering pack before the bots arrived.

How did you get sixty stakeholders to one log? I made the log the only place a date lived. If a dependency was not a row with an owner and a date, it did not exist for the program. People stopped sending emails because the log was where decisions happened.

What decision did you challenge? The original plan had robots at 100% from day one with a two-week "stabilization." I challenged the word stabilization — it had no number — and replaced it with the gate and the dial.

How did you handle the freeze? Someone always has an exception. One exception path, written down: a change in freeze needed a customer-facing defect it fixed, a rollback, and my signature. Three got through in ten weeks. The freeze held because the exception path was real and narrow.

RESULTS PROBES

How did you measure +15%? Units processed per direct labor hour, inbound and outbound, versus the two buildings' combined trailing baseline for the same weeks, normalized for the dotcom-versus-store mix. Weekly, from WMS and the labor system, in the steering pack.

How did you measure pick accuracy? WMS pick-confirm scan versus gold-standard location, by zone, by hour. A building average would have hidden the bad zones inside the good ones — which is exactly how 94% was first noticed.

Compared to your goal? Zero incidents: met. 98% by zone before full volume: met, one week late versus the vendor's own plan. Efficiency above baseline: exceeded.

What trade-off did you make? Three weeks of robotics ROI, deliberately, for zero customer risk through peak. I would make it again.

I'm concerned a 94% print means your testing was weak. Tell me more. Fair. The zone logic passed test volume and failed real volume mix — the test data under-represented multi-unit orders in the dense zones. The miss was in the test design. The mechanism caught it in week two instead of at peak, which is what the gate was for; and the fix I made since is below.

CHALLENGE

What would you do differently? Shadow the bots on one zone for a week at full real volume before they touched any customer order. I now put "one zone, real volume, a week" into every automation ramp I write.

Why this story for Highest Standards? Because the standard was set before it was tested, and it cost us something to hold it. Standards that are free are not standards.

At Amazon. A launch or retrofit ramp as a dial with zone- or path-level gates: the capacity plan counts a new building's or retrofit's throughput only for the process paths that have cleared the rate and quality bar at real volume. You never plan on capacity that has not earned it.

S2 — Cyber Monday, 4 AM: rebalancing allocation across three DCs with 70% of the information (Manhattan Associates)

LPs: Bias for Action · Are Right, A Lot · Dive Deep · Customer Obsession. **AIQB:** Bias for Action 2, 4, 5 · Are Right, A Lot 1, 3 · Dive Deep 2 · Customer Obsession 8.

Problem Peak day, 4 AM: backorder lines piling up across a 3-DC / 100-store network. Millions of dollars of orders would miss cutoff. Root cause unknown; full diagnostic an hour away.

Goal Hold fill rate at 99% through the peak day — which meant stopping backorder growth inside the hour.

Key metric Fill rate by hour (target 99%); backorder lines; orders at risk of missing promised cutoff (in \$).

Did it move? **Yes.** Fill rate held at 99% through peak; ~\$8M of at-risk sales shipped to promise; structural stockouts ~20% over the engagement once the triage logic became standing policy.

Situation. "I owned order allocation and promising across three mega-DCs and about a hundred stores. The node we sourced from determined the lead time the customer saw — choose the node, and you have chosen the promise. Cyber Monday, 4 AM, a VP on the phone: backorder lines climbing, millions at risk, and the full diagnostic was an hour out.

Task. One number: hold fill rate at 99% through the day. That meant stopping the backorder growth inside the hour, with about seventy percent of the information I wanted.

Action. I split what had to be decided now from what could be learned in parallel. On the bridge I ranked every open order by *promise-at-risk* — dollar value times hours to its cutoff — instead of by age, and rebalanced allocation across the three DCs to the nodes that had hours, overriding the engine's assignments manually in that priority order. Every decision timestamped in one log, so the diagnostic team could later separate my effects from the fault's. I kept two actions explicitly off the table: no blanket promise downgrade, and no emergency inbound — both one-way doors at 4 AM. The trace an hour later showed the allocation scheduler had died quietly at the daylight-saving change. Inventory was fine. DC capacity was fine. The engine had simply stopped assigning.

Result. Fill rate held at 99% through the peak. Roughly eight million dollars of orders that would have missed cutoff shipped to promise. The promise-at-risk ranking became the standing allocation priority, and structural stockouts were down twenty percent over the engagement. Afterward I wrote a scheduler heartbeat and a DST checklist into peak readiness. I never want a second 4 AM."

Supply · Demand · Capacity. **Demand:** peak-day orders by promise tier and cutoff. **Supply:** inventory by DC and store — adequate, as it turned out. **Capacity:** two kinds — the physical pick/pack hours by node, which were fine, and the *planning-system* capacity to assign orders to nodes, which had gone to zero. **What bound:** a false capacity signal. The lesson for a capacity planner: when the plan says "capacity is short," first prove the shortage is physical.

What was innovative. The standard move at 4 AM is FIFO — clear the oldest backorders first — and a blanket promise downgrade to stop the bleeding. I ranked by dollar-weighted hours-to-cutoff so the scarce decision bandwidth protected the most promise per minute, and I classified every available action as a one-way or two-way door before taking it. The timestamped log turned a firefight into a dataset.

SITUATION PROBES

Why was this important? Peak day is a disproportionate share of the year's revenue, and a missed cutoff on peak is a customer who does not come back. The VP's phrase was "millions by noon."

What if you had waited for the diagnostic? An hour of growing backorders at peak order rates. We modeled it after the fact: the at-risk pool roughly doubled every forty minutes at that hour's order arrival rate.

What did you know at 4 AM, and what didn't you? Knew: backorder lines, inventory by node, DC hours scheduled. Didn't know: why the engine was not assigning. I decided I did not need the "why" to move orders safely by hand, as long as every move was logged and reversible.

ACTION PROBES

What did you do versus the ops team? I took the bridge, set the ranking rule, made the node assignments in priority order, and kept the log. Ops executed the picks; the engineers ran the trace. The ranking rule was mine.

Why act at 70%? What would have made you wait? Waiting was a customer miss with a dollar sign; acting was a two-way door — manual assignments could be re-run when the engine came back. I would have waited if the action were a one-way door: a promise downgrade or an emergency PO. Those I kept off the table.

Why promise-at-risk and not FIFO? FIFO protects the oldest order; it does not protect the most promise per minute of my attention. With an hour of hand assignment, I wanted every minute to clear the most dollar-hours of risk. FIFO would have spent the hour on low-value orders with late cutoffs.

How did you decide which DC got which order? Hours available at the node against the order's cutoff, then inventory at the node, then distance to the customer — in that order. If a node had the units but not the hours before

cutoff, it did not get the order.

What was the risk of the manual rebalance? Double allocation when the engine resumed. The timestamped log was the mitigation: when the scheduler came back, the engineers reconciled against the log before re-enabling auto-assignment.

What decision did you challenge? The first instinct on the call was to downgrade all two-day promises to standard. I refused — that is a one-way door that touches every customer to fix a problem that might be local. It was local.

RESULTS PROBES

How did you compute \$8M? Orders that were in the backorder pool at 4 AM and shipped before their cutoff after the rebalance, at order value. Conservative: it does not count orders that would have joined the pool over the next hour.

How did you measure fill rate? Lines shipped before cutoff ÷ lines ordered, by hour, against the 99% line. Hourly, because a daily average would have hidden a 4 AM hole.

How did the -20% in stockouts come about? That's not from one night. Correct. The ranking rule became the standing allocation priority — scarce units went to the orders with the highest promise-at-risk rather than first-come. Over the engagement, structural stockouts fell twenty percent against the prior-year baseline. The night was the proof of concept.

Compared to goal? Goal was 99% through the day. Met. We did not claw back the backorders that had already missed before 4 AM; I counted those as a miss and said so.

CHALLENGE

What would you do differently? The heartbeat and the DST check existed by the following week. The deeper fix: I now treat every peak as a different operating regime with its own readiness checklist, because the failure modes at peak are not the failure modes at average.

Why this story for Bias for Action and not something more recent? Because it is the cleanest case where speed had a price and I can show the math on both sides. The Sephora robot throttle (S1) is more recent and is the same muscle: a two-way door turned the same day.

At Amazon. When a region shows a capacity shortfall, I would first prove it is physical — hours or cube — before spending a lever. And every lever in the plan would be labeled one-way or two-way before peak, so the bridge at 4 AM already knows which ones it may pull without a meeting.

S3 — Two hundred lanes: plan lead time at the empirical p99, release the inventory the brochure was holding (Elementum)

LPs: Think Big · Ownership · Dive Deep · Customer Obsession · Frugality · Invent and Simplify. **AIQB:** Think Big 1, 3, 6 · Ownership 3 · Dive Deep 5, 6 · Frugality 1 · Invent and Simplify 1, 6.

Problem	The inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured. Safety stock on every lane was sized against a number nobody had checked; planners padded it by feel.
Goal	Planned lead times that hold 99% of the time, <i>and</i> less safety stock, not more — measured variability should buy inventory out, not in.
Key metric	Lead-time accuracy (arrivals inside planned lead time, by lane, by week) → 99%; safety stock units and dollars released, auditable by lane.
Did it move?	Yes. 99% lead-time accuracy held. 580,000 units and ~\$1.5M of safety stock released. Drift flag per lane made it permanent.

Situation. "A planner asked me for a lead-time variability report. When I looked underneath, the inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured against an actual arrival. Planners

knew the number was soft, so they padded safety stock by feel. Nobody owned the seam between the brochure and the plan.

Task. The ask was a report. The goal I set was different: planning lead times that hold ninety-nine percent of the time, with *less* safety stock — because if you measure variability honestly, it should release inventory on most lanes, not add it.

Action. I pulled the shipment history across two hundred lanes myself — ship date, arrival date, carrier, and the source of the transit number, brochure or scan. I built the empirical distribution per lane: mean, spread, tail. Then the move that mattered: I set every planning lead time to the lane's empirical p99 — the smallest number of days by which 99 of 100 shipments had already arrived — not the mean, and not a normal approximation, because right-skewed lanes have fatter tails than a normal distribution admits. The counter-intuitive result: on most lanes the measured p99 was tighter than brochure-plus-gut, so stock came out; on the minority where it was wider, stock went in, deliberately. I wrote the release rule with finance before any dollar moved: stock released only where measured spread was tighter than assumed. Then I productized it — a drift flag per lane that fires when the rolling distribution widens past a threshold, so widening triggers a review instead of a silent re-plan.

Result. Ninety-nine percent lead-time accuracy held. Five hundred eighty thousand units and about a million and a half dollars of safety stock released, auditable lane by lane. The ask was a report. The product was lane-level planning as a system — and the drift flag is why it was still true a year later."

Supply · Demand · Capacity. **Supply:** inbound lead time per lane — the variable I measured. **Demand:** replenishment requirements by SKU-location, unchanged. **Capacity:** safety stock is storage capacity and working capital consumed by supply uncertainty. **What bound:** a bad supply parameter was consuming cube and cash. Fixing the parameter released both without touching a building. This is the cheapest capacity there is: capacity hidden inside a wrong assumption.

What was innovative. The standard answer is a variability report and a safety-stock formula with a normal z-score. I planned to the empirical tail per lane, wrote a release rule that required the measurement to be *better* than the assumption before money moved, and built a drift detector so the plan re-opened itself. The system replaced the report.

SITUATION PROBES

Why important? What if nothing happened? Every lane carried either too much stock (cash and cube) or too little (stockouts), and nobody could say which. Doing nothing meant continuing to pay for an unmeasured number in inventory every week.

What was the initial scope and how did it change? One report for one planner. I expanded it to all two hundred lanes and to the planning parameter itself — I took that to leadership with mockups before I built it, because changing the planning parameter is a change to the plan, not to a report.

What was the goal before you set yours? There was none beyond "a report." That was the problem.

ACTION PROBES

Why p99 and not p95 or the mean? The mean is a coin flip — in a right-skewed lane, roughly two in five shipments arrive after it. p95 leaves one shipment in twenty late, which at our order frequency was a stockout a month on a busy lane. 99% was the service level the business had already committed to; I made the lead time honor it. On lanes where the cost of carrying justified less, the method allows a lower quantile per lane — the point is to choose it, not inherit it.

Why empirical quantile and not a normal distribution? Transit times are right-skewed — a shipment can be very late but not very early. A normal fit understates the tail, which is exactly where the stockouts live. I checked: on long ocean lanes the normal p99 was days short of the empirical p99.

How much history per lane? Seasonality? Enough shipments per lane to make a p99 meaningful — thin lanes were pooled with similar lanes (same origin region, mode, carrier) and flagged as lower confidence. Peak-season transits were kept as their own regime where the data supported it; the drift flag catches the seasonal widening on lanes where it did not.

What did you do versus the team? I pulled the data, built the distributions, chose the quantile rule, pitched leadership, wrote the release rule with finance, and designed the drift flag. Engineering productized the pipeline from my spec.

How did you get planners to trust a number that cut their safety stock? I showed each planner their own lanes: brochure number, measured p99, actual late arrivals under each. Where the data said "add stock," I added it first. Credibility came from the lanes where I gave inventory back.

What did you challenge? The assumption that the carrier's number was a fact. And the plan to ship a report — a report would have been read once.

RESULTS PROBES

How did you measure 99%? Actual arrivals inside the planned lead time, by lane, by week, from scan data — the same data the distributions came from, held out forward in time.

How did you compute \$1.5M? Safety-stock units before and after, by SKU-lane, valued at standard cost. Released only on lanes where the measured spread was tighter than the assumed spread — the finance-signed rule — so no lane's release depends on a judgment call.

Was the whole 580K units a release? Did any lanes get more stock? The 580K is net. A minority of lanes got more stock because their measured tail was wider than the brochure. I would rather not quote the lane count without the file in front of me.

Trade-off? Lanes with wider tails carried more inventory than before. That was the right trade: we had been running them short and calling it luck.

Compared to goal? 99% accuracy: met. Less safety stock, not more: met, net \$1.5M. Permanence: drift flag live.

CHALLENGE

What would you do differently? Start every new domain with a measurement audit in week one — "which planning parameters are measured, and which are inherited?" — instead of discovering the brochure problem mid-project. I have done that on every program since, including S/4HANA.

Why this story for Think Big? Because the opportunity was bigger than the ask, I took it, and I can show the gap: a report versus a planning system that still runs.

At Amazon. Every capacity planning parameter — dock-to-stow hours, process-path rates, vendor lead times, days of cover — is a distribution, not a number. I would ask on day one which of them are planned at a mean, plan the ones that bind at a chosen quantile, and put a drift flag on each so the plan re-opens when the world moves.

S4 — Forecast alerts that entered the weekly S&OP because planners trusted them (Elementum)

LPs: Invent and Simplify · Deliver Results · Earn Trust · Dive Deep. **AIQB:** Invent and Simplify 3, 5 · Earn Trust 1 · Dive Deep 6 · Customer Obsession 3, 4.

Problem	Planners learned about shortages when a production line stopped. Forecast accuracy ~70% at SKU-location-week. The SAP data any alert would run on was not trusted — the ERP's own reports disagreed with each other.
Goal	Alerts planners act on inside the weekly S&OP; forecast accuracy 70% → 85%; safety stock reduced — in that order, because the third is legitimate only after the first two.
Key metric	Forecast accuracy at SKU-location-week; daily reconciliation drift (MD04 vs MB52); alert false-alarm rate; safety stock \$ released.
Did it move?	Yes. 70% → 85%. Zero recon drift daily. ~\$2M safety stock released under a two-signal rule. Live in three months; adopted into S&OP without mandate.

Situation. "Planners found out about shortages when a line stopped. Forecast accuracy was about seventy percent at SKU-location-week, and the SAP data behind any alert was not trusted — the demand view and the stock view disagreed on any given day.

Task. Alerts planners would act on inside the weekly S&OP, forecast accuracy to eighty-five, and safety stock down. In that order.

Action. I refused to ship a threshold email on data nobody trusted. I led the SAP ingestion — inventory, production, POs, ASNs, forecasts — into one supply-and-demand view, with a daily MD04-to-MB52 reconciliation published and held at zero drift, and one rule: no alert fires on a day the recon is red. The alert itself was coverage math — projected supply against demand by week — with a thirty-second reason line: which input moved, by how much. Then a public false-alarm burn-down: every alert a planner dismissed was classified and the logic fixed in a weekly forum until the dismiss rate fell. The forecast-accuracy gain came mostly from the inputs the recon exposed — stale POs, duplicated demand, wrong lead times — not from a new model.

Result. Live in three months. Forecast accuracy from seventy to eighty-five percent at SKU-location-week. Recon at zero drift daily. About two million dollars of safety stock released — only where *both* forecast error and lead-time error had improved, a rule finance signed. The alert entered the weekly S&OP because planners trusted it, not because anyone mandated it."

Supply · Demand · Capacity. Demand: the forecast at SKU-location-week — the input I made 15 points more accurate. **Supply:** inventory, production, open POs, ASNs — reconciled into one view. **Capacity:** two kinds — the S&OP's capacity to act (alert lead time before a shortage), and safety stock as storage/working-capital consumed by forecast error. **What bound:** trust. Without a reconciled base, no amount of alerting moves a decision.

What was innovative. The standard approach is a dashboard with thresholds. I built the trust mechanism first — a daily reconciliation with a hard "no alert on a red day" rule — then alerts that explain themselves in one line, then a public burn-down of false alarms, and a two-signal release rule so no single improved number could justify a dollar. The alert was the least important part.

SITUATION PROBES

Why important? Cost of nothing? A stopped line is the most expensive shortage there is, and planners were compensating with safety stock they could not justify. Doing nothing was paying for 30% forecast error in inventory and in expedites.

Who was the customer? The planners, and through them the plants. The S&OP meeting was where their decisions were made, so that was where the alert had to earn a seat.

ACTION PROBES

Why reconciliation first? Planners wanted alerts. An alert on untrusted data trains people to ignore alerts. I would rather be three weeks late with something they believe than a day early with something they dismiss. I told them that in the first meeting.

What's the MD04-to-MB52 reconciliation in plain terms? The planning view of stock and requirements versus the physical stock view, by plant, by material, by day. If they disagree, one of them is wrong, and any alert built on either is wrong. We held the difference at zero and published it.

Why a reason line? A planner has thirty seconds per alert. "Coverage falls below two weeks in week 7 because PO 4471 slipped nine days" is actionable. "Coverage alert" is noise.

How did you run the false-alarm burn-down? Every dismissed alert got a category — wrong data, wrong logic, right alert wrong timing, right alert ignored — and we fixed the top category weekly in a public forum. The dismiss rate was on the wall. Planners saw their feedback change the logic within a week; that is where trust came from.

How did forecast accuracy improve 15 points if you did not change the model? Garbage in. Stale POs were being counted as supply; duplicated demand from two systems inflated requirements; lead times were brochure numbers. The recon exposed each one. Fixing inputs moved the number more than a new model would have.

What did you personally own? The ingestion scope and the recon design, the "no alert on a red day" rule, the reason-line requirement, the burn-down forum, and the two-signal release rule with finance.

RESULTS PROBES

How did you measure forecast accuracy? $1 - |\text{forecast} - \text{actual}| / \text{actual}$ at SKU-location-week, lag-matched to the planning horizon, weekly, in the S&OP pack.

Why two signals before releasing stock? Forecast accuracy alone could be mix; lead-time accuracy alone could be a quiet quarter. Requiring both to improve on the same SKU-location is how I kept the \$2M honest. Finance signed the rule before the first dollar.

Compared to goal? Adoption: met, un-mandated. 85%: met. Stock release: \$2M, met under the stricter rule.

Trade-off? Three weeks of calendar on the recon before anyone saw an alert. Worth it; the alternative was alerts nobody read.

CHALLENGE

Differently? Put the reason line in the first prototype — I added it after planners dismissed the first week's alerts, and I lost a week of trust I had to earn back.

At Amazon. Any capacity signal I bring to S&OP or Finance carries its reconciliation state and a one-line reason — "FC-A storage utilization crosses 90% in week 9 because inbound for category X pulled forward two weeks." And a lever fires on two signals, not one: forecast and placement, or utilization and dock-to-stow.

S5 — Austria: I arrived certain it was carriers; the data said we could not measure lateness (Elementum)

LPs: Dive Deep · Are Right, A Lot · Earn Trust · Have Backbone. **AIQB:** Are Right, A Lot 1, 6 · Dive Deep 1, 2, 4 · Earn Trust 6 · Have Backbone 2.

Problem A 25-location European freight program: reports said carriers were late everywhere and freight cost was over plan. The sponsor had committed to a routing engine to move volume to better carriers.

Goal Auditable freight savings through better routing — the sponsor's number — without building an engine on data that could not support a recommendation.

Key metric Data quality at the planning grain (reported 40%; true ~55%) → 99% by defect class; freight savings \$ auditable to a shipment.

Did it move? **Yes.** Data quality 99% by defect class before any routing ran. ~\$8M freight savings across 25 locations, every dollar auditable to a shipment under a finance-signed method.

Situation. "I walked into a 25-location European freight program in Austria convinced we had a carrier problem. Every report said carriers were late everywhere, freight cost was over plan, and the sponsor had promised a routing engine to move volume to better carriers.

Task. Deliver auditable freight savings. The sponsor wanted the engine first.

Action. Before the engine kickoff I pulled three years of shipment history myself — not an analyst, me. The data said something more embarrassing than late carriers: we could not measure lateness. True data quality at the planning grain was about fifty-five percent, against the forty the reports claimed — wrong in both directions because the reports were measuring different things. Missing rates, lanes keyed at the wrong grain, the same carrier under five spellings, transit times copied from brochures. So I changed the program's first milestone from engine to measurement, in front of a sponsor who had promised the engine, with one page: a routing recommendation on 55% data is a coin flip with a confident interface. We fixed quality by defect class to ninety-nine percent before any routing

ran. And I had finance sign the savings method — baseline lane rate versus executed rate, per shipment — before anyone claimed a dollar.

Result. About eight million dollars in freight savings across 25 locations, every dollar traceable to a shipment. Data quality ninety-nine percent by defect class. And a changed mind, on the record: I had been hired to fix carriers, and I told the people who hired me that carriers were not the problem."

Supply · Demand · Capacity. **Supply:** carrier capacity and rates by lane. **Demand:** shipment volume by lane across 25 locations. **Capacity:** the program's capacity to decide — you cannot route what you cannot measure. **What bound:** measurement, not carriers. For a capacity planner the lesson is literal: a utilization or on-time number at the wrong grain is wrong in both directions, and spending a lever on it is a coin flip.

What was innovative. The usual response to "the data is bad" is to clean it on the way into the engine. I made measurement the deliverable, scored by defect class with a 99% bar, and signed the savings method with finance before the first claim — so the result could not be argued later. I also changed my own thesis in public, which is what made the sponsor willing to change theirs.

SITUATION PROBES

Why important? What if nothing happened? An engine on 55% data would have moved volume to carriers that looked good because their data was missing. We would have claimed savings nobody could audit and lost the sponsor's credibility with finance.

What was the initial scope? Routing engine, 25 locations, savings target. I kept the target and the locations and inserted a measurement phase in front.

ACTION PROBES

Why pull it yourself? Because I was the one who was going to tell the sponsor his first milestone was wrong. I needed to have seen the rows.

How can "40% reported" and "55% true" both be wrong? The reports counted a field as present if it had any value; some values were brochure numbers, so the reports overstated quality on those. Elsewhere they counted lanes as incomplete because of spelling variants that were actually resolvable, so they understated. Measured at the planning grain with defect classes, the true number was 55% — and the point is that neither report number could be used for anything.

How did you move a sponsor who had promised the engine? One page, privately first: here is what the engine would recommend on three real lanes using current data, and here is what actually happened on those lanes. Two of three recommendations were wrong. He took the page to his leadership himself.

What did you do versus the team? I pulled the history, defined the defect classes and the 99% bar, re-sequenced the program, and negotiated the savings method with finance. The team built the cleansing by class.

What decision did you challenge? My own first. Then the sponsor's milestone. Then the finance assumption that savings could be claimed on a lane average — we went per shipment.

RESULTS PROBES

How did you compute \$8M? Per shipment: baseline rate for that lane and mode before the program, minus executed rate after routing, summed across 25 locations, under the method finance signed. No lane averages.

Why 99% and not "good enough"? Because the engine's recommendations are only as reliable as the worst defect class. At 99% by class, a wrong recommendation was rare enough to catch by exception. At 90% it was a weekly argument.

Compared to goal? Savings target: met and auditable. The engine shipped later than the sponsor had promised. I owned that date change in writing.

CHALLENGE

Differently? Pull the data in week one, before the kickoff deck, not week four. It would have saved the sponsor an embarrassing re-plan.

Why this story for Are Right, A Lot? Because being right here meant changing my mind in public with data, which is the only version of that principle that is hard.

At Amazon. Before I plan to a utilization number, a design rate, or an on-time metric, I would ask what grain it is measured at and how it was last validated against actuals. A capacity ceiling that was set three years ago and never re-measured is a brochure number.

S6 — Seven to sixty-eight customers behind six graduation gates; cost-to-serve down 95% while the book grew ten times (Flexport)

LPs: Insist on the Highest Standards · Deliver Results · Frugality · Success and Scale Bring Broad Responsibility. **AIQB:** Highest Standards 1, 6 · Deliver Results 3 · Frugality 1, 8 · Success and Scale 1, 8.

Problem A global 4PL shipment-visibility platform with seven pilot customers, cost-to-serve \$0.66 per container, a support team already stretched, and a sales pool of 60+ waiting to board. Scaled the usual way, support cost scales with customers.

Goal 68 customers by the committed date, with cost-to-serve *down* and ticket response *up* at volume.

Key metric Cost-to-serve per container (\$0.66 baseline); ticket response rate (72%); customers live vs plan; data quality and ETA accuracy per wave.

Did it move? **Yes.** 68 customers, two months early. Cost-to-serve \$0.66 → \$0.03 (−95%) while the base grew ~10×. Ticket response 72% → 90% at 5× volume.

Situation. "A global 4PL shipment-visibility platform: seven pilot customers, sixty-six cents a container to serve, a support team already stretched, and sixty-plus customers in the sales pipeline. Scaled the way pilots usually scale, support cost would have grown with the customer count.

Task. Sixty-eight customers by the date — with cost-to-serve going down and ticket response going up at volume. Both, or the scale was not worth having.

Action. I treated onboarding as a capacity problem. I wrote six graduation criteria a wave had to clear before it boarded, published to sales and leadership: data quality at or above ninety percent against source of truth, milestone and ETA accuracy at or above eighty, verification SLA, engagement floors, platform stability, CSAT. A wave was admitted only when the data and support layer had cleared the bar at the *current* load — so each wave paid for the next by forcing the fixes that made the next one cheaper. I held wave three in front of everyone on an ETA-accuracy miss traced to one bad data source, while sales wanted the pool boarded. We fixed the source, built a kill switch for that class of source, and boarded a week later.

Result. Sixty-eight customers, two months early. Cost-to-serve from sixty-six cents to three cents a container — a ninety-five percent cut — while the base grew about ten times. Ticket response from seventy-two to ninety percent at five times the volume. Unit cost falling while the book grows only happens if the gates keep bad data out."

Supply · Demand · Capacity. **Demand:** customer onboarding — the sales pool, arriving faster than the platform could absorb. **Supply:** carrier and port data feeds — the raw material; bad feeds were the defect. **Capacity:** the support and data layer's ability to serve customers, measured as cost-to-serve per container. **What bound:** capacity to serve at quality. The gates metered demand to capacity and forced capacity to grow cheaper than demand.

What was innovative. The usual scaling plan is a headcount curve. I inverted it: admission was gated on the platform's measured ability to serve at the current load, so scale was earned, wave by wave, and the cost curve bent down instead of up. The public hold on wave three was the mechanism teaching itself.

SITUATION PROBES

Why important? Cost of nothing? Boarding sixty customers onto a platform that cost sixty-six cents a container would have been a profitable product turned into a support organization. The pilot economics did not survive multiplication.

Who set the 68 and the date? Sales and leadership. I accepted both and added the two conditions — cost down, response up — so "68" could not be met by burning the team.

ACTION PROBES

Why those six criteria? Each was a leading indicator of a support ticket: bad data → a ticket; bad ETA → a ticket; low engagement → churn; instability → a ticket; CSAT → the lagging check. If a wave cleared all six, it generated fewer tickets per customer than the previous wave. That is the whole cost curve.

Why hold wave three publicly? Sales was furious. A public hold teaches the mechanism; a private hold teaches politics. Sales saw the criteria, saw the miss, saw the fix, saw the wave board a week later. Nobody argued a gate after that.

What did you do versus the team? I wrote and published the criteria; I made the hold call and named the data source; I decided the kill switch was a product requirement, not a support workaround. Engineering built the switch; support ran the waves.

What decision did you challenge? The plan to hire support ahead of the curve. I said: hire after the gate proves what the next wave costs. We hired less than planned.

How did you set priorities with sixty customers waiting? Waves were sequenced by data-source quality, not by deal size — the customers whose carriers had clean feeds boarded first and made the platform cheaper for the harder ones.

RESULTS PROBES

How did you compute cost-to-serve? Fully loaded support and data-ops cost for the platform ÷ containers tracked, monthly. \$0.66 at seven customers, \$0.03 at sixty-eight.

Isn't a 95% drop mostly denominator — ten times the containers? Partly, and I say so. Ten times the containers alone would have given about \$0.07 if cost had stayed flat. It did not stay flat — it fell, because tickets per customer fell wave over wave. The rest of the drop is the gates.

How did ticket response improve at 5× volume? Fewer tickets per customer (gates), plus the kill switch removing a whole class of noisy tickets, plus the verification SLA making response a measured commitment.

Compared to goal? 68: met, two months early. Cost down: −95%. Response up: 72 → 90. All three.

Trade-off? One week on wave three, and a hard conversation with sales. Cheap.

CHALLENGE

Differently? Build the kill switch into the platform before wave one. I knew bad sources existed; I let the first miss force the fix.

At Amazon. New capacity — a launch, a retrofit, an overflow site — graduates into the plan behind published criteria at real volume, and the plan counts only graduated capacity. And the cost side: I would want the capacity plan's cost per unit to fall as the network grows, which only happens if each wave of capacity is admitted at a quality bar rather than a date.

S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table (Flexport)

LPs: Have Backbone; Disagree and Commit · Think Big · Frugality · Earn Trust. **AIQB:** Have Backbone 1, 2, 6, 8 · Think Big 2, 3 · Frugality 4 · Earn Trust 3.

Problem An inherited eighteen-month, operator-first launch sequence, already socialized with leadership. Engineering capacity was fixed; the sequence decided when value arrived and what it cost to serve.

Goal Value to customers sooner, at lower total cost-to-serve, with the same engineering headcount.

Key metric Engineering effort to first value; time to value (months); cost-to-serve reduction (%); cost avoided (\$).

Did it move? Yes. Engineering effort -66%. Time to value 6 months vs 18. Cost-to-serve cut ~50% vs ~25% under the original sequence. ~\$5M cost avoidance.

Situation. "I inherited an eighteen-month, operator-first launch plan that leadership had already seen and nodded at. Engineering capacity was fixed. The only real variable was sequence: which half of the problem we solved first.

Task. Get value to customers sooner at lower cost-to-serve, without new headcount — and do it without a public fight that would make my manager choose between me and the plan he had presented.

Action. I costed both sequences in the same currency — cost-to-serve, the number the business was actually paid on — rather than in features. Partner-first cut total cost-to-serve by about fifty percent and shipped value in six months; operator-first cut about twenty-five percent and shipped in eighteen. Two columns, same rows, same assumptions, every assumption footnoted. I did not flip it in a meeting. I took the table to my manager privately and asked him to break it. He pressure-tested it with engineering for two days; it held; and the Bellevue onsite ratified the new sequence. Then I committed completely — the operator work I had deprioritized, I personally protected a date for so the operators were not the losers.

Result. Sixty-six percent less engineering effort to first value. Value in six months instead of eighteen. About five million dollars of cost avoidance. I did not win the argument with seniority. I won it with a table — and the table is why it stayed won."

Supply · Demand · Capacity. **Capacity:** engineering hours — fixed, and the scarce resource. **Demand:** two customer populations, partners and operators, both wanting the platform. **Supply:** the sequence in which fixed capacity was allocated to demand. **What bound:** capacity allocation. The flip was a capacity-planning decision in a different costume: same hours, re-sequenced against value per hour.

What was innovative. Roadmaps are usually argued in features and dates. I converted the argument into the business's own P&L currency, made the two options structurally identical so only the sequence differed, and chose the private route so the decision-maker could change his mind without losing face. Backbone is the private table; theater is the public ambush.

SITUATION PROBES

Why important? Eighteen months of fixed engineering capacity is the most expensive thing the business owned. Spending it in the wrong order was a cost-to-serve decision nobody had priced.

Cost of nothing? Twelve extra months before customers saw value, and a cost-to-serve reduction half the size. The plan was not wrong; it was unpriced.

How socialized was it? Presented to leadership, dates in the planning system, teams staffed to it. Flipping it meant my manager going back to the people he had presented to.

ACTION PROBES

Why not raise it in the planning meeting? Because a public challenge forces the person who owns the plan to defend it, and then the argument is about him, not the number. Privately, he could attack the table and own the change. He did both.

What if he had said no? I would have asked what number would change his mind, gone to get it, and committed to operator-first in the meantime without relitigating. Disagree once, with data; then commit.

Why not do both in parallel? We did not have the engineers. Parallel would have shipped neither in six months. Sequencing was the whole bet.

How confident were you in the ~50% vs ~25%? Confident in the ratio, less in the absolutes — I said so in the footnotes. Engineering's two-day pressure test moved a few assumptions and the ratio held.

What did you do versus the team? I built the table, chose the currency, chose the private route, and protected the operators' date afterward. Engineering validated the effort estimates.

What decision did you challenge? The plan itself, and the implicit assumption that a socialized plan is a finished plan.

RESULTS PROBES

How did you compute \$5M? Difference in projected cost-to-serve between the two sequences over the planning horizon, plus the engineering effort not spent on the operator-first path to first value. Conservative assumptions, footnoted, reviewed by engineering and finance.

What did the operators lose? Twelve months of waiting for their screens. I owned that trade and protected a date for it. They got a worse year and a better platform.

Compared to goal? Sooner: 6 vs 18. Cheaper: -66% effort, ~50% CTS cut. Same headcount: yes.

CHALLENGE

Differently? Price the customer-side cost explicitly — speed, breach risk — not only cost-to-serve. The table was right; it would have been more right with a third row.

Why this story for Backbone? Because it was a real disagreement with the person who wrote my review, about a plan with his name on it, and I can show the artifact that resolved it.

At Amazon. When Finance says a building is not justified, I do not argue — I price both columns in the same currency: annualized cost of the build against overflow, transfers, premium transportation, speed downgrades and the risk-adjusted cost of a peak miss, with the break-even forecast shown. The six-page narrative this JD asks for is that table with sentences around it.

S8 — Four systems into one pipeline: the weekly review is only as good as the data under it (Flexport)

LPs: Invent and Simplify · Dive Deep · Frugality · Ownership. **AIQB:** Invent and Simplify 1 · Dive Deep 5 · Frugality 5, 6 · Ownership 2.

Problem The Monday business review ran on four systems stitched together in Excel — by me. Two hours a week, ~60% system accuracy, one person as the single point of failure, and a VP steering on numbers that disagreed with their sources.

Goal A review whose data is boring: accuracy $\geq 90\%$, zero manual assembly, no single point of failure — without a new team or a new tool budget.

Key metric Metric accuracy vs source systems (60%); manual hours per week (2); cost of status quo ($>\$100\text{K}/\text{yr}$ priced); parallel-run delta ($<1\%$).

Did it move? **Yes.** Accuracy 60% $\rightarrow$ 90%. Manual hours $\rightarrow$ 0. Parallel run within 1% for consecutive weeks before cutover. First automated catch: a 4% ticket-sync drop the manual era never saw.

Situation. "Flexport's Monday review ran on four systems stitched together in Excel by me — two hours every week, about sixty percent agreement with the source systems, and a VP making decisions on it. I was the single point of failure, and I knew the numbers were soft.

Task. Make the data under the review boring: ninety percent accuracy or better, no manual assembly, no dependence on me — and do it without a new team.

Action. I wrote the BRD myself and froze seven metric definitions exactly as the team already used them — no new metrics, no renamed ones, because changing the vocabulary during a migration lets everyone relitigate every red. I priced doing nothing at over a hundred thousand dollars a year in the infrastructure team's own labor rates, so the request competed on their numbers, not mine. I gated cutover on a parallel run: manual versus pipeline within one percent for consecutive weeks, published. Four systems became one Snowflake pipeline. I hit one wall I did not plan

for — ticket data carries customer content, and the security review cost two weeks. I owned it upward the day I found it and re-sequenced; the parallel run meant Monday never missed.

Result. Accuracy from sixty to ninety percent. Two manual hours a week to zero. The first automated catch was a four percent ticket-sync drop the manual era had never seen, because the manual era could not see it. The weekly cadence only works if the data underneath it is boring."

Supply · Demand · Capacity. Demand: decisions the review had to support every Monday. **Supply:** data from four systems. **Capacity:** the review's capacity to see — bounded by two manual hours and one person. **What bound:** the review itself. A WBR is a capacity system for decisions; this story is about expanding its throughput and removing its single point of failure.

What was innovative. Most "automate the dashboard" projects redesign the metrics at the same time and lose the room. I froze the vocabulary as a contract, made the status quo compete on the infrastructure team's own cost, and treated cutover like a WMS go-live — a numeric parallel-run gate. The pipeline was the easy part.

SITUATION PROBES

Why important? A VP was steering on numbers that were forty percent wrong somewhere. One Monday the wrong forty percent would be the one that mattered.

Cost of nothing? Two hours a week of my time, priced; the latent cost of a bad decision, unpriced; and a review that stopped the week I was out.

ACTION PROBES

Why freeze the definitions? Some were bad. Some were. I wrote the bad ones down for a later change with its own approval. Changing definitions mid-migration means nobody can tell a pipeline bug from a definition change, and every red becomes an argument about the chart instead of the business.

Why price the status quo in their rates? Because the infrastructure team had to build it, and "it saves me two hours" is not a priority for them. "It costs you \$100K a year of your own people's time" is.

Why a 1% parallel run — why not just switch? Because the review was live. The parallel run is the mock cutover: I would not go live on a WMS without one and I would not go live on the VP's numbers without one.

What went wrong and what did you do? I did not plan the data-security review; ticket text carries customer content. Two weeks. I told my manager the day I found it, with the new date and the reason, and the manual path kept Monday alive. Every data program since has "whose approval does this data need, and what is its lead time" on the kickoff checklist.

What did you do versus the team? BRD, frozen definitions, status-quo pricing, parallel-run gate, the escalation. Infrastructure built the pipeline.

RESULTS PROBES

How did you measure 60% and 90%? Each of the seven metrics, weekly, recomputed from source systems and compared to the review value. Share of metric-weeks within tolerance.

The 4% catch — why does it matter? Ticket sync had been silently dropping records, which made response rates look better than they were. The manual process could not detect a drop it was built on top of. The first week of the pipeline, it flagged. That is the difference between a dashboard and a mechanism.

Compared to goal? 90%: met. Zero manual: met. No SPOF: met. Two weeks late: owned.

CHALLENGE

Differently? Security review on the kickoff checklist. I have never missed it since.

At Amazon. Before I change a single metric in a capacity WBR I would freeze the existing definitions, run the new source in parallel against the old for a published number of weeks, and only then cut over. New metrics come after, through their own door.

S9 — Dual-DC WMS migration in eight months, tolerance published before mock one (Sephora JDA)

LPs: Learn and Be Curious · Deliver Results · Ownership · Earn Trust. **AIQB:** Learn and Be Curious 1, 2 · Deliver Results 1, 2 · Ownership 2 · Earn Trust 4.

Problem	Two DCs on a legacy ERP had to move to JDA WMS in eight months. A \$20M ROI pipeline of four programs was gated behind it. Nothing was documented, and I had never run a WMS migration.
Goal	Both DCs live on time and on the \$1.8M budget, with inventory reconciliation within 0.5% at cutover and no disruption to store or dotcom replenishment.
Key metric	Inventory reconciliation miss at each mock cutover vs the 0.5% tolerance; schedule; budget; hypercare exit criteria.
Did it move?	Yes. Mocks: 7% → 1.8% → 0.3% vs 0.5%. On time. ~3% under the \$1.8M budget. \$20M pipeline unblocked. One exit criterion slipped 1.5 weeks — owned, below.

Situation. "Sephora handed me a dual-DC migration from a legacy ERP to JDA WMS on a hard eight-month clock. Four programs worth about twenty million in ROI were gated behind it. Nothing was documented, and I had never run a WMS migration.

Task. Both DCs live on time and on a \$1.8 million budget, with inventory reconciled within half a percent at cutover, and stores and dotcom never noticing.

Action. I did not fake experience. I sat the sponsor for objectives, assumptions and success criteria in week one, and held discovery's written exit until every flow had a documented current state — that was the gate I could run without domain expertise, and it forced the expertise into the room. Then entrance and exit criteria for every phase. The decision I am proudest of: I published the 0.5% inventory reconciliation tolerance, agreed with finance, before mock one — so the first mock would be a measurement, not a verdict. Three mocks: seven percent, then 1.8, then 0.3. The seven went up unvarnished with my name on it. The go/no-go read the scorecard, not the mood.

Result. On time. About three percent under budget. 0.3% against a 0.5% tolerance at cutover. Four-week hypercare with numeric exit criteria. Twenty million of downstream ROI unblocked. One miss: training sign-off, a go-live exit criterion, slipped a week and a half — I had scored the RF-gun conversion as low effort, my own pre-training test caught the upgrade script failing on newer hardware, and COVID night shifts killed the recovery window. I told leadership it was mine. The program still landed on the date."

Supply · Demand · Capacity. **Demand:** daily store and dotcom replenishment that could not stop. **Supply:** inventory accuracy — the reconciliation tolerance is a supply-truth metric. **Capacity:** two DCs' ability to ship through a cutover weekend. **What bound:** the one-way door. Cutover is capacity you take offline on purpose; the mocks were how I bought it back with a number.

What was innovative. Publishing the tolerance before the first mock inverts the usual politics: a 7% first result is data, not failure, and the trend to 0.3% is the story. And running discovery as a gate I could hold without expertise turned my inexperience into a forcing function for documentation.

SITUATION PROBES

Why important? Four programs and \$20M of ROI were waiting. Every month of slip was a month of four other teams idle.

What was the biggest risk? A cutover that reconciled badly and had to roll back — a week of two DCs shipping from a system of record nobody trusted.

ACTION PROBES

You'd never done this. Why not hand it to a specialist? I used specialists for the WMS. I owned the gates, the tolerance, and the go/no-go — which is the part that needs an owner, not an expert.

How did you learn fast enough? Discovery was my classroom. Holding the exit until every flow was documented meant the experts had to teach the program what the building did. By mock one I could read a reconciliation report.

Why publish the tolerance before mock one? So the first number could not be argued. If 7% arrives and the tolerance is set afterward, the tolerance becomes 7%.

Why 0.5%? Finance's materiality for inventory at the two buildings. I did not pick it; I got them to sign it.

What did you do about the 7%? Read it as a scorecard: classified every miss by cause — mapping, timing, physical — and burned the top class before mock two. 1.8% was mostly timing; 0.3% was noise.

What did you do versus the team? I set the tolerance and criteria, held the gates, made the go/no-go, and owned the slip. The integrator built; ops counted.

RESULTS PROBES

Compared to goal? On time: met. Budget: 3% under. Tolerance: 0.3 vs 0.5. Training exit: 1.5 weeks late. Three of four, and the fourth owned in writing.

Tell me more about the slip — it's a go-live exit criterion you missed. A single-point dependency — RF guns — in front of an exit criterion, scored low effort because it was "just a script." The script failed on newer hardware. The rule since: any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimated size. It has caught two since.

CHALLENGE

Differently? The what-if rule above, from day one.

Why this story for Learn and Be Curious? Because the learning had a mechanism — discovery as a gate — and a measured outcome, not a book list.

At Amazon. Any one-way-door capacity change — a building cutover, a system migration, a network re-architecture — gets its tolerance published before the first rehearsal, and the rehearsals are read as a trend, not a verdict.

S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback (Sephora S/4HANA ↔ ISM)

LPs: Ownership · Invent and Simplify · Earn Trust · Dive Deep · Deliver Results · Are Right, A Lot (being wrong).

AIQB: Ownership 1, 3 · Invent and Simplify 1, 3 · Earn Trust 1, 6 · Are Right, A Lot 4, 5 · Deliver Results 2.

Problem S/4HANA was becoming the inventory system of record for stores, retail DC and dotcom with a warehouse execution system beside it. "Available" meant something different in every system because each definition protected a different number. Customer returns, RTV stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.

Goal One signed definition per stock state, with one owner and one promising rule; returns handled as first-class inventory; cutover with zero customer-facing incidents and a rollback that is a number, not a paragraph; the legacy inventory system (MCS) retired.

Key metric Units promised against non-sellable states (returns, QI, blocked, in-transit) → zero; customer-facing incidents at go-live; interface sync and p95 latency vs rollback triggers; MCS retired.

Did it move? **Yes.** Non-sellable promises → zero, verified in production. Zero customer-facing incidents on staged go-lives. 343-task runbook; rollback triggers never fired. MCS retired. One rule I wrote was wrong and was corrected through the same signed policy.

Situation. "Sephora was standing up S/4HANA as the inventory system of record for stores, the retail DC and dotcom, with a warehouse execution system beside it. The problem was not the software. 'Available' meant something different in every system, because each definition protected a different number — fill rate for stores, labor for the DC,

the promise for dotcom, the credit memo for finance. Customer returns, return-to-vendor stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.

Task. One definition per stock state, one owner, one rule about who may promise against it — signed by the business before anyone configured anything. Returns as first-class inventory. A cutover with zero customer-facing incidents. And retire the legacy inventory system.

Action. I redesigned the inventory operating model. Every stock state — available-to-sell, reserved, in-transit, quality-inspection, blocked, lost-and-found — got one definition, one owner, and one promising rule, and I took the whole set to steering with the failure mode attached to each: here is the customer who gets hurt if this state is promised. Returns got a plant of their own. Return-to-vendor stock-transport orders were modeled as pick-reserve so credit could never run ahead of the physical unit: receive, grade, then restock, scrap or RTV — nothing promiseable until disposition. I ran it as a program: RAID log, a weekly CIO/COO pack across sixty-plus stakeholders on frozen metrics so a red stayed red, and a 343-task weekend cutover runbook with rollback written as a number before the weekend — interface sync under 99% for four hours, or p95 latency over twice budget for one hour. One rule I wrote was wrong: I forbade every channel from promising against units on a truck. It starved dotcom of sellable units while in-transit inventory rose. The data showed the miss; the fix went through the same signed policy — a per-lane in-transit promise window from measured lane data.

Result. Could a unit in quality-hold or on a returns pallet still be promised to a customer? No — verified in production. Zero customer-facing incidents on the staged go-lives. The rollback never fired. MCS came out only after the new definition held in production. And the in-transit correction recovered the sellable units I had hidden, through policy, not a hotfix."

Supply · Demand · Capacity. **Supply:** inventory by state — the point is that only some states are supply. **Demand:** three channels promising against the same pool. **Capacity:** returns are inbound volume the plan did not forecast; quality-hold is storage you cannot sell; in-transit is supply you may or may not count. **What bound:** definitions. A capacity plan that counts non-sellable states as available overstates usable capacity, and the promise lies. Giving each state a definition and an owner is the same move as giving each building type its own ceiling.

What was innovative. Treating "available" as a policy document signed by the business before configuration, with the failure mode written next to each state; modeling RTV as pick-reserve so finance could not out-run the floor; and writing rollback as two numbers so a tired bridge at 2 AM could call it without a meeting. Also: correcting my own wrong rule through the policy, not around it.

SITUATION PROBES

Why important? Cost of nothing? Every oversell was a cancelled order and a customer who did not trust the site; every return promised before grading was a damaged unit shipped to a second customer. Finance was issuing credit on units the floor had not touched.

What was the initial scope? Configure S/4HANA to match the legacy definitions. I changed it to: define first, configure second, because the legacy definitions were the problem.

ACTION PROBES

Why a signed policy before configuration? Most oversell bugs are vocabulary bugs. A config change nobody signed becomes next quarter's argument. Once the sentence was signed, the design arguments got small — the plant and the STOs were just the implementation of a sentence everyone had agreed to.

Why a separate returns plant? That's heavy. Because a returned unit is not the same supply as a new one until someone has graded it, and the only way to make the system enforce that was to make it physically and logically elsewhere. The heaviness is the point: nothing in that plant is promiseable until it leaves.

Why pick-reserve for RTV? So finance's credit memo could not run ahead of the physical unit. The reservation ties the money to the pick. Before, credit was issued on paper and the unit was sometimes still on a store shelf.

Why rollback as a number? A rollback that is a paragraph gets debated at 2 AM by tired people. A number gets called by whoever is on the bridge. The triggers existed so the weekend could start.

How did you keep sixty stakeholders from re-litigating reds? Frozen metrics in the weekly pack. A red could not be made green by changing the chart; it could only be made green by moving the number.

What did you do versus the team? Wrote the stock-state definitions and the failure modes; designed the returns plant and the pick-reserve RTV model; owned the RAID, the pack, the runbook, and the rollback triggers; made the MCS retirement call. SAP and the integrator configured.

RESULTS PROBES

How did you verify "non-sellable promises → zero"? Promise log joined to stock state at promise time, daily in hypercare. The count of promises against QI, blocked, returns-plant or lost-and-found stock: zero.

How did you know the in-transit rule was wrong? Sellable units by channel fell while units in transit by lane rose. Two lines diverging is the signal. Fix: a per-lane in-transit promise window from measured lane transit — the 200-lane idea applied to promising instead of planning.

What did the wrong rule cost? Weeks of dotcom running with fewer sellable units than it physically had. I will not quote a dollar I did not measure; I can say the recovery was visible in sellable units the week the window went live.

Compared to goal? Definitions signed: yes. Returns first-class: yes. Zero incidents: yes. Rollback never fired: yes. MCS off: yes. One rule wrong and corrected: owned.

CHALLENGE

Differently? Set the in-transit window from lane data on day one instead of starting from "never." Conservatism has a price, paid in the same currency as risk. And sample the OMS-to-WMS handshake daily in hypercare from day one.

At Amazon. Returns are inbound volume the plan did not forecast; quality-hold is cube you cannot sell; in-transit is supply you may count only inside a measured window. I would want the capacity plan's "available" to be a signed definition by stock state — and I would want to know which states today's utilization number includes.

S11 — The \$2.5M discrepancy I refused to fix quietly (Manhattan Associates)

LPs: Earn Trust · Customer Obsession · Have Backbone. **AIQB:** Earn Trust 6 · Customer Obsession 1, 8 · Have Backbone 1.

Problem An OMS-to-WMS handshake defect had mis-stated a couple hundred customer orders — roughly \$2.5M of order value — and the manager's instinct was to correct the data quietly.

Goal Protect the customer's real priorities and the account, with zero undisclosed corrections.

Key metric Orders corrected to the customer's priority; disclosure before fix (yes/no); account outcome.

Did it move? **Yes.** Full disclosure; the customer prioritized the orders that mattered to their business; the class was bulk-patched; we later won a larger initiative with them.

Situation. "On an OMS-to-WMS handshake we found a discrepancy across a couple hundred orders, roughly two and a half million dollars of order value. **Task.** The manager's instinct was to correct it quietly and move on. I thought the customer had to decide the order of repair, because they knew which orders mattered. **Action.** I argued for full transparency and won it: we showed the customer the list the same day, let them prioritize, and bulk-patched the defect class rather than the instances — so the fix was structural, not cosmetic. It was uncomfortable; I was the junior voice in that room. **Result.** They protected the orders that mattered to their business, the class never recurred, and we later won a larger initiative with them. Silent corrections are loans against trust, and the interest rate is brutal. Same instinct at Flexport later: a truncated week printed 99.4% automation and I discarded it, because a VP steering on three data points is a decision-quality problem."

Supply · Demand · Capacity. **Demand:** the customer's orders, mis-stated. **Supply:** the truth about them. **Capacity:** the account's trust — the capacity to be believed next time. **What bound:** the willingness to show a bad number.

Why not fix it and tell them after? Same outcome. Not the same outcome: they would have learned we fix things without telling them. Also not the same repair — they re-ordered the fix by business priority, which we could not have known.

What did the manager say? That it would create noise. I agreed it would, and said the noise now was cheaper than the noise later. He let me make the call and own it.

Why bulk-patch the class, not the instances? Two hundred orders were the symptom. The handshake was the cause. Fixing instances would have made the next two hundred our fault twice.

At Amazon. If the capacity plan is wrong, the WBR hears it from me first with the decomposition attached, before anyone finds it in a speed miss. Reds that arrive pre-decomposed are the version of this story that scales.

S12 — Giving the skeptic the gate (Sephora agile conversion)

LPs: Hire and Develop the Best · Earn Trust · Strive to be Earth's Best Employer. **AIQB:** Hire and Develop 1, 3, 4 · Earth's Best Employer 6, 9 · Earn Trust 3.

Problem A VP sponsored converting a twenty-person product/engineering/QA group from a six-week to a four-week release. The QA lead — the most experienced person in the room — was the most skeptical, and QA was where a faster cycle would break first.

Goal Four-week releases without a quality regression; the mechanism survives my leaving.

Key metric Release cycle (6 → 4 weeks); critical defects per release; whether the gate was still in use after I rolled off.

Did it move? **Yes.** Four-week cycle reached in six weeks. Critical defects –50%. The QA-authored gate was adopted by neighboring teams and outlived my engagement.

Situation. "A VP sponsored moving a twenty-person product, engineering and QA group from six-week to four-week releases. The QA lead was the most skeptical person in the room — and QA was where a faster cycle breaks first.

Task. Four-week releases, no quality regression, and a mechanism that would survive me leaving. **Action.** Instead of managing around her, I gave her the load-bearing artifact: the definition of done with entry criteria, and the authority to hold a release that did not meet it. I coached her on writing a gate that can say no — numeric, published, owned by her — and on how to defend it to engineering without me in the room. Within two cycles she was the one defending it. **Result.** Four-week cycle in six weeks. Critical defects dropped by half, because the gate was stricter than what had existed at six weeks. Neighboring teams copied the model and the ceremony survived my leaving. Development that depends on me staying is not development."

Supply · Demand · Capacity. **Demand:** releases per quarter, up 50%. **Capacity:** the team's throughput at quality. **Supply:** the gate — the mechanism that let throughput rise without quality falling. **What bound:** trust in the person who would hold the line.

Why the skeptic? Why not the person most bought in? The skeptic knew where it would break. Giving her the gate converted her objections into criteria, and her credibility made the gate real to engineering.

What specifically did you coach? How to write a criterion as a number, not an adjective; how to hold a release in a room full of engineers who want to ship; and when to let something through with a documented exception rather than block. She was good at the first two already; the third was the coaching.

What's the honest shape of your people leadership? Direct reports in the classic sense at MetricStream — onshore/offshore delivery teams, SOWs, the go-live chair. Everywhere since, matrix: this twenty-person conversion,

UDC with 15+ tracks and 60+ stakeholders, about a hundred people across nine teams sequenced at Flexport. I do not dress matrix leadership up as a headcount I did not hold.

Did an investment in someone not work out? At Flexport I spent a quarter coaching an analyst toward owning the Monday review; he did not want the visibility and said so late. I should have asked earlier what he wanted rather than what I needed. I ask that question in the first month now.

At Amazon. The person who most doubts the capacity plan — in Ops, in Finance — is the person I would ask to write the gate the plan has to clear. Their objections become the criteria, and the plan is stronger for it.

S13 — Two go-lives, one engineering team, no 24-hour days (Sephora)

LPs: Strive to be Earth's Best Employer · Deliver Results · Invent and Simplify. **AIQB:** Earth's Best Employer 3, 10 · Deliver Results 1, 7.

Problem Two programs compressed onto one engineering team with overlapping go-lives; roughly \$200K per month per DC at stake in delay, and a team that would otherwise work around the clock.

Goal Both go-lives on date, with issue resolution moving around the clock and no individual working around the clock.

Key metric Go-live dates met; issue age in the RAID log; hours worked per person per week (held to normal shifts).

Did it move? Yes. Both landed on date; ~\$1M per DC protected; no 24-hour days.

Situation. "Two programs compressed onto one engineering team, with roughly two hundred thousand dollars a month per DC at stake in any slip. The default was heroics — the same six people on every call, day and night. **Task.** Both go-lives on date without burning the team. **Action.** One RAID log across both programs, so an issue existed once and had one owner. One 45-minute daily forum, not two. And a two-pass war room: a day shift that triaged and assigned, a night shift in another time zone that executed and handed back — so issues moved around the clock without anyone working around the clock. I set the rule publicly: no one on both passes. **Result.** Both landed on date, about a million dollars per DC protected, and the issue-age metric in the log fell because work never waited for morning. Heroics are a process miss."

Supply · Demand · Capacity. **Demand:** two programs' issues. **Capacity:** one team's hours — fixed. **Supply:** the two-pass structure that made the same hours cover twice the clock. **What bound:** hours per day, not people.

How did you enforce "no one on both passes"? The RAID log recorded who touched what and when. Anyone appearing on both passes got a conversation with me, not a thank-you.

How did you measure the \$1M per DC? Monthly cost of delay per DC — roughly \$200K — times the months of slip the original single-pass plan projected. Conservative: it ignores the morale cost of the alternative.

What if the night pass hit something only the day pass could decide? Published escalation: one named person per night with decision rights for anything reversible; anything irreversible waited for morning by design. That boundary was written down before the first night.

At Amazon. Peak readiness is the same shape: a 24-hour bridge staffed in passes with written decision rights, so the 4 AM call has an owner who is awake and a boundary they do not cross alone.

S14 — Teaching myself a stack: the LLM triage agent (independent build)

LPs: Learn and Be Curious · Frugality · Insist on the Highest Standards. **AIQB:** Learn and Be Curious 2, 5, 7 · Frugality 5 · Highest Standards 2.

Problem Incident triage cost ~\$25 per incident in analyst time; I did not know the stack to automate it.

Goal Automate triage at a fraction of the cost with accuracy held and every conclusion cited to a source.

Key metric Cost per incident; triage accuracy vs the evaluation set; citation coverage.

Did it move? Yes. \$25 → \$0.40 per incident (−98%); accuracy held against the pre-written evaluation set; 100% citation coverage.

Situation. "Incident triage cost about twenty-five dollars an incident in analyst time. I wanted to automate it and did not know the stack. **Task.** Build it, learn the stack, and hold accuracy — not ship a demo. **Action.** Python, FastAPI, tool use. The discipline I brought from supply chain: I wrote the evaluation set before the agent — a hundred-plus real incidents with known-correct triage — and every prompt or model change was gated on the known past still passing. Every output had to cite its source or it did not ship. **Result.** Cost per incident from twenty-five dollars to forty cents — a ninety-eight percent cut — accuracy held against the evaluation set, one hundred percent citation coverage. Same operating system as everything else I run: instrument first, then let the scorecard teach."

Use only if asked about learning something new or about AI. It is a side build, not a program you owned at a company. Say that in the first sentence.

How did you learn the stack? By building against a scorecard. Documentation plus a failing evaluation set is the fastest teacher I know.

Why the evaluation set first? Because otherwise "it works" means "it worked on the three I tried." The set is the gate; the agent is the candidate.

At Amazon. Any model or tool I bring to capacity planning gets a backtest against last year's weeks before it touches the plan. The evaluation set is the mock cutover.

S15 — A scope cut that was priced, published, and scored (Flexport FF App export launch)

LPs: Deliver Results · Bias for Action · Customer Obsession · Frugality. **AIQB:** Deliver Results 1, 4 · Bias for Action 1 · Frugality 2, 4.

Problem An export-side launch for forwarding partners with a hard date and a scope that could not all ship: Air and LCL, operator screens, and the partner path competed for the same engineers.

Goal Ship the MVP on date; eliminate the most monthly operator interactions per engineering hour; migrate existing import partners without churn.

Key metric Monthly operator interactions eliminated (the success metric set before scope); partners live; migration rate of import partners.

Did it move? Yes. MVP April 2022; 11 partners, 10 countries; 48 of 53 import partners migrated within a month. Air/LCL and operator screens deferred — openly, with dates.

Situation. "The export launch was minimum-scope by design: partner path first; Air and LCL waited a year; operators kept their old screens two more quarters. Real cuts, real complaints. **Task.** Ship on date and make every cut defensible. **Action.** I set the success metric before scope — monthly operator interactions eliminated, not features shipped — so every proposed cut was scored against that number, in a published table, not in a hallway. Then a burn-up tripwire: any new work was estimated and swapped for something of equal size, never absorbed. **Result.** MVP landed April 2022 — eleven partners, ten countries — and forty-eight of fifty-three import partners migrated within a month. The people who lost scope could read why in the table."

Supply · Demand · Capacity. **Capacity:** engineering hours to the date — fixed. **Demand:** three constituencies' features. **Supply:** the scored scope. **What bound:** hours; the metric decided who got them.

Why interactions eliminated and not revenue? Revenue lagged by quarters and was attributable to many things; operator interactions were measurable monthly and were the cost we were paid to remove.

How did you keep the door shut after launch? The burn-up chart was public; any line added had to show the line removed. Twice someone tried to absorb work; both times the chart made it visible within a week.

At Amazon. A capacity plan's scope is its lever list. Every lever added after the plan is published gets a cost and swaps for one of equal cost — or the plan is a wish list.

S16 — A thousand routes, one gate (Elementum EDI)

LPs: Insist on the Highest Standards · Deliver Results · Dive Deep. **AIQB:** Highest Standards 1, 3 · Deliver Results 3 · Dive Deep 5.

Problem ~1,000 supplier and carrier routes to onboard onto X12 EDI with SAP PO and ASN integration; shipment completeness at 60%, so planners could not see what was coming.

Goal All routes live with completeness $\geq 95\%$; no cohort graduates on bad data.

Key metric Valid ASN rate per cohort (gate $\geq 95\%$); shipment completeness (60% baseline).

Did it move? **Yes.** Completeness 60% → 95%. Every cohort graduated at $\geq 95\%$ valid ASNs. Routing-guide overrides carried reason codes.

Situation. "About a thousand supplier and carrier routes had to go onto X12 EDI with SAP PO and ASN integration; shipment completeness sat at sixty percent, so the plan was blind to forty percent of what was in motion. **Task.** All routes live, completeness to ninety-five, and no cohort allowed through on bad data. **Action.** Cohorts graduated only at ninety-five percent or better valid ASNs — one gate, published, no exceptions by relationship. And a detail that mattered: the routing-guide skeptics were often right about specific lanes, so I made overrides legitimate — every override carried a reason code, which turned sabotage into data and gave me the lane list to fix. **Result.** Shipment completeness from sixty to ninety-five percent. Every cohort graduated at the bar. The reason codes became the backlog for the routing guide's next version."

Supply · Demand · Capacity. **Supply:** inbound shipments — visible only if the ASN is valid. **Demand:** planners' need to see supply in motion. **Capacity:** inbound planning capacity is bounded by visibility — you cannot schedule a dock for a shipment you cannot see. **What bound:** data completeness.

Why 95% and not 100%? The last five percent were edge formats that cost more to fix than the visibility they bought; I documented them as a known residual with an owner instead of pretending the gate was 100%.

Why let people override the routing guide? Because they were going to anyway. A reason code made the override visible and often correct; forbidding it would have made it invisible and always suspicious.

At Amazon. Inbound capacity planning depends on ASN accuracy and appointment adherence; I would gate vendor cohorts on measured ASN validity and treat every dock-appointment override as a reason-coded data point about where the plan is wrong.

6. Leadership Principles — which story, which hook

They will not say the LP's name. They will say "tell me about a time." Hear the LP, pick the row, speak the hook, tell the story, stop.

LP	LEAD / BACKUP	THE HOOK SENTENCE
Customer Obsession	S3 / S1	"I treated the brochure as a lie the customer was paying for in inventory." / "I will not ship a customer-facing miss to protect a robotics narrative."
Ownership	S3 / S10	"Nobody owned the seam between brochure lead times and the plan. I took the seam, not the ticket." / "Nobody owned what 'available' meant. I wrote the definitions, took them to steering with the failure mode attached, and owned the cutover."
Invent and Simplify	S8 / S10	"New metrics without killing old ones is complexity wearing a badge. I froze seven definitions and killed the spreadsheet." / "Most oversell bugs are vocabulary bugs. One definition, one owner, one rule per stock state — then the design arguments got small."
Are Right, A Lot	S2 / S10	"70% is enough when waiting is a customer miss." / "I forbade promising against in-transit units. I was wrong, the data said so, and the fix went through the same signed policy."
Learn and Be Curious	S9 / S14	"I had never run a WMS migration. I did not fake it. I built the gate and used the specialists."
Hire and Develop the Best	S12	"I gave the skeptic the gate. Development that depends on me staying is not development."
Insist on the Highest Standards	S6 / S1	"I held wave three in front of everyone. Standards without a stop-the-line are posters."
Think Big	S3 / S7	"The ask was a report. The product was lane-level planning as a system." / "Think Big without sequencing is a slogan."
Bias for Action	S2 / S1	"Timestamped decision, then a post-mortem that makes silence an alarm." / "Throttling the robots was a two-way door we had built. I turned it the same day." <i>One-way doors (S/4HANA cutover, JDA cutover) get rehearsals and a numeric rollback instead.</i>
Frugality	S6 / S7	"Cost-to-serve down 95% while the book grew ten times. Headcount would have scaled the waste."
Earn Trust	S11 / S10	"Silent corrections are loans against trust." / "Sixty stakeholders, one weekly pack, metrics frozen so nobody could re-litigate a red by changing the chart."
Dive Deep	S5 / S8	"I pull until the unit is a shipment, an order, a lane or a building — not a national average."
Have Backbone; Disagree and Commit	S7	"Backbone is the private table. Theater is the public ambush."
Deliver Results	S9 / S10	"Results with a rollback are not results. The gates are why it counted." / "343-task runbook, rollback as a number, never fired, legacy system off, zero customer incidents."
Strive to be Earth's Best Employer	S13 / S12	"Heroics are a process miss."
Success and Scale Bring Broad Responsibility	S6	"We scaled ten times and ticket response improved. The cost of scale was paid in gates, not in someone else's backlog."

Question-bank index — hear the question, know the story

Amazon's interviewer question bank gives 6–11 suggested questions per Leadership Principle; interviewers pick two or three. The most likely ones for an L6 supply-chain loop, in the bank's own wording (abridged), and the story written for each. If you hear a question not listed, find the LP, pick the lead story in the table above.

LP	QUESTION FROM THE BANK (ABRIDGED)	STORY
Are Right, A Lot	1. Didn't have enough data to make the right decision. What path did you take? Was it correct?	S2, S5
	4 / 5. A bad decision or error in judgment. Impact? What did you learn and apply?	S10 (in-transit rule), S9 (RF guns)
	6. Discovered your idea was not the best course of action. How did you find out? What was?	S5
Bias for Action	1. A calculated risk where speed was critical. How did you mitigate?	S1, S15
	2 / 4 / 5. Tight deadline, no time for all options / respond immediately / move forward vs gather more.	S2
	8. Removed a serious roadblock preventing your team from progressing. (Manager)	S8 (security review), S13
Customer Obsession	3 / 4. Anticipated a customer need / used customer feedback to drive improvement.	S4, S3
	8. Balanced the needs of the customer with the needs of the business.	S1, S11
Deliver Results	1. Delivered an important project under a tight deadline. What sacrifices?	S9, S15
	2. Significant unanticipated obstacles to a key goal.	S1 (94%), S9, S10
	3. Not only met a goal but considerably exceeded it.	S6
Dive Deep	1 / 2 / 4. Complex problem; dig to root cause; a big problem you helped solve — what information was missing?	S5, S2
	5 / 6. A metric you used — or created — to identify a need for change.	S3 (p99), S4 (recon drift), S8
	7. Validated the assumptions underlying a report's project. (Manager)	S5
Earn Trust	1. Communicated a change in direction people would have concerns with.	S4, S10
	4. Not able to meet a commitment. Impact? Learned?	S9 (training slip), S8
	6. Uncovered a significant problem in your team. How did you communicate it?	S11, S5
Frugality	1. Helped save costs or eliminate waste.	S3, S6
	5 / 6. Creative solution without additional resources / requested funding — did you try another approach?	S8, S13

LP	QUESTION FROM THE BANK (ABRIDGED)	STORY
Have Backbone	1 / 2. Strongly disagreed with manager or peer / unpopular stance in front of your leader.	S7, S5
	8. The business gained because you persisted.	S7, S3
Hire and Develop	1 / 3 / 4. Helped a team member develop; gave feedback; coached.	S12
Highest Standards	1 / 4. Refused to compromise standards / chose between standards and delivery.	S1, S6
	6. Most significant continuous-improvement project you led.	S6, S16
Invent and Simplify	1. Complex problem solved with a simple solution.	S8, S10
	3 / 5. Made something simpler for customers / usual approach wouldn't work.	S4, S3
Learn and Be Curious	1 / 2. Needed deeper expertise / took on work outside your comfort area.	S9, S14
Ownership	2. Didn't think you'd meet a commitment. How did you identify and communicate the risk?	S8, S9
	3. Initiative that benefited the company but wasn't any group's responsibility.	S3, S10
Think Big	1 / 3. Saw an opportunity to do something much bigger / proposed a novel approach.	S3
	2. Changed the direction of a function and helped them embrace new thinking.	S7
Earth's Best Employer	3 / 6 / 10. Improved the work experience / empowered someone / improved the team's environment.	S13, S12
Success and Scale	1 / 8. Impact beyond your immediate customer / left something better than you found it.	S6, S12

The follow-up discipline

- **"What did you do, specifically?"** One sentence that starts with "I." Name the artifact: the gate, the table, the log, the SQL.
- **"Why not [the obvious alternative]?"** Name the cost of the alternative in the same currency as the result.
- **"How did you measure?"** Metric, grain, baseline, target. Never "we saw improvement."
- **"What would you do differently?"** A real change you have since made. Not "nothing" and not "communicate more."
- **"Walk me through the data pull."** Tables joined, grain, filters. If you did not pull it yourself, say who did and what you checked.

7A. How Amazon manages capacity — the unit, the flow, the money, the review

This is the depth section. You will not recite it. It exists so that when the interviewer says "walk me through how you would think about capacity in a region," every step, metric and dollar you name is attached to a unit moving through a building. **Three kinds of statements appear here.** (1) Structure that Amazon has described publicly — FC types, inbound and outbound process paths, sort centers and delivery stations, CPT. Use freely. (2) Standard capacity-planning economics — utilization ceilings, fixed-cost dilution, peak factors, labor math. Use freely; this is the job. (3) Numbers. Every number below is **ILLUSTRATIVE**, chosen so the arithmetic can be checked on paper. Say so if you use one, and say what replaces it on day one: the S&OP demand file, Ops Engineering design rates, Finance's cost-per-unit ledger.

7A.0 Supply, demand, capacity — three words, four equations

Say these three words in this order, with these meanings, and the interviewer knows you are speaking the team's language. **Demand** is units customers will order, by week, by region, by building type, by speed tier — the forecast, with its error. **Supply** is units arriving to be stowed — vendor inbound, FBA inbound, transfers, returns — and the inventory already on hand. **Capacity** is what a building can do with those units: **store** them (cube) and **move** them (hours through each process path). Demand and supply are flows; inventory is the stock between them; capacity is the pipe and the tank. The plan is the statement that for every building and every week the four equations below hold.

- (1) INVENTORY BALANCE
$$I(t) = I(t-1) + \text{Supply_in}(t) - \text{Demand_out}(t) - \text{Shrink}(t)$$

what you stow this week minus what you ship is the change in what you hold
- (2) STORAGE CONSTRAINT (Ceiling_storage $\approx$ 0.85–0.90)
$$I(t) \times \text{cube_per_unit}(\text{mix}) \leq \text{UsableCube} \times \text{Ceiling_storage}$$

what you hold must fit — at the ceiling, not at 100%
- (3) INBOUND THROUGHPUT (steps: unload, receive, decant, stow)
$$\text{Supply_in}(t) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(t) \times \text{Rate}_s]$$

you can only stow what every inbound step can push in the hours you have
- (4) OUTBOUND THROUGHPUT (steps: pick, rebin, pack, SLAM, ship-sort)
$$\text{Demand_out}(t, \text{CPT}) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(\text{window before CPT}) \times \text{Rate}_s]$$

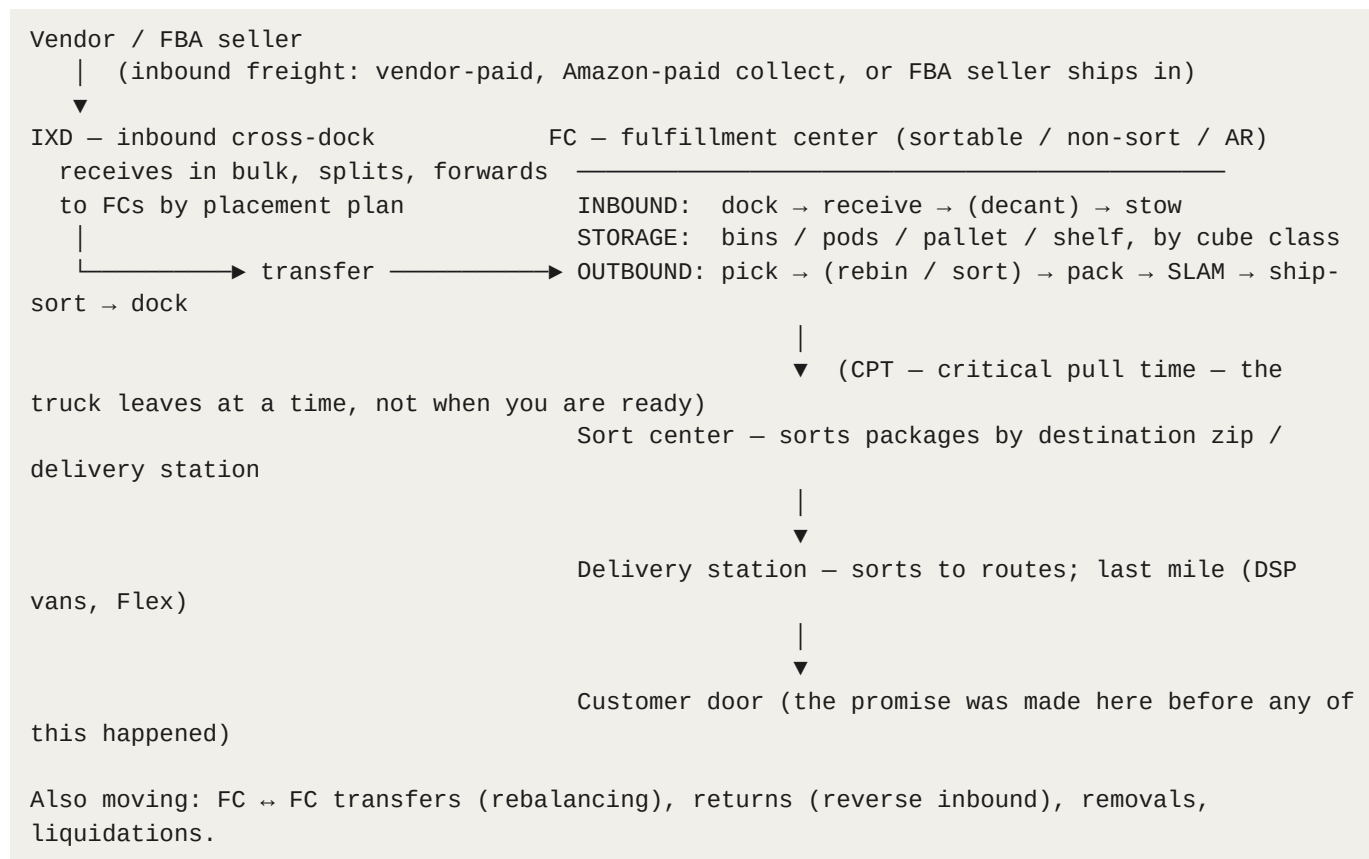
you can only ship what every outbound step can push BEFORE EACH TRUCK LEAVES
- Rate_s is not a constant:
$$\text{Rate}_s = \text{DesignRate}_s \times f(\text{utilization}) \times g(\text{new-hire share}) \times h(\text{flow evenness})$$

f falls past ~85–90% storage utilization (walk, hunt, congestion)
 $g \approx 0.6 \rightarrow 1.0$ over ~2 weeks of ramp; $h < 1$ when work arrives in bursts

How to read a gap with these. If (2) fails, you have a *storage* gap: lower $I(t)$ (days of cover, slow movers out, inbound pushed back) or raise UsableCube (overflow, retrofit, new build). If (3) fails, *inbound* gap: hours, appointment shaping, IXD absorbs. If (4) fails, *outbound* gap: hours, placement to buildings with slack, CPT structure. If (1) is drifting up week over week with (2) approaching the ceiling, you have a *placement or forecast* problem wearing a capacity costume — supply is being sent to a building whose demand does not justify it. Name the equation that failed before you name a lever. That is the whole discipline.

The one sentence: "Demand tells me how many units must leave each building each week; supply tells me how many must arrive; inventory is the difference; capacity is whether the difference fits in the cube and whether the flows fit in the hours — before each truck leaves."

7A.1 The network one unit travels



The sentence that organizes everything below: capacity is planned in two currencies — **cube** (can the unit be stored) and **hours** (can the unit be moved through each step before its truck leaves) — and the plan is honest only if both are checked at the step where each binds. A building can have empty shelves and still miss CPT because pack is short of hours. A building can have hours and still refuse inbound because it is 97% full.

7A.2 Inbound — step by step, what the unit does, how it is measured, what it costs

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Inbound transportation	Trailer or container travels vendor → IXD/FC. Appointment booked; trailer arrives at a dock door.	Trailers/day vs appointments; on-time arrival; units per trailer (cube fill); dwell in yard (hours).	Dock doors × turns/day; yard spots; appointment slots.	Variable: linehaul \$/mile or \$/trailer, drayage, fuel; semi-fixed: yard jockeys, gate staff. Paid by Amazon on collect freight; embedded in cost of goods on prepaid.
Dock / unload	Pallets or fluid-loaded cartons come off. Cartons scanned to a receive queue.	Cartons unloaded/hour; door dwell (hours door is occupied per trailer); unload backlog (trailers waiting).	Doors; unload labor hours; forklift/equipment.	Variable labor: unload hours × rate. Fixed: doors, levelers, forklifts (capex / lease).

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Receive	Carton opened, ASIN identified, quantity counted, condition checked, unit now exists in inventory as "received, not yet stowed."	Units received/hour per associate (receive rate); receive backlog (units or hours of work waiting); vendor compliance defects (mislabeled, wrong qty) per 1,000 cartons.	Receive hours; problem-solve queue for exceptions.	Variable labor: receive hours. Exceptions cost 5–10× a clean carton (illustrative) — the compliance metric is a cost metric.
Decant / prep	Units pulled from vendor cartons into totes or prepped (poly-bag, label) so they can be stowed.	Units/hour; prep required %; tote fill.	Prep hours; consumables.	Variable: labor + packaging consumables.
Stow	Unit placed into a bin (shelf, pod in AR, pallet location). Inventory becomes available to promise at this scan.	Units stowed/hour (stow rate); dock-to-stow hours (the headline inbound flow metric: carton arrival to unit available); stow backlog; bin-fullness at stow.	Stow hours; empty cube — if bins are full, stow slows or stops; in AR, pod availability and robot drive-unit count.	Variable labor: stow hours. Stow rate falls as the building fills (walk time, hunting for empty bins) — this is the first place the utilization curve costs money.

Why dock-to-stow is the inbound metric that matters. Until the unit is stowed, the customer cannot buy it against this building's promise, the vendor's trailer is still at your door or in your yard, and the carton is occupying floor space that is not a bin. A dock-to-stow target (say ≤ 24 hours, illustrative) converts inbound capacity into a flow commitment. Miss it and you have an inbound backlog, which shows up as refused appointments, which shows up as inventory stuck upstream, which shows up as a speed miss three weeks later.

7A.3 Storage — the cube currency

QUANTITY	HOW TO COMPUTE IT (ILLUSTRATIVE)	WHAT IT TELLS YOU
Gross storage cube	Bins $\times$ cubic feet per bin, by bin type (shelf, pod, pallet, bulk floor).	The building's theoretical ceiling. Never plan to it.
Usable cube	Gross $\times$ bin-fill efficiency (air inside a bin because units are odd shapes; ~60–75% illustrative) $\times$ operating ceiling (85–90%).	The number you actually plan storage to.
Units on hand	Daily outbound units $\times$ days of cover (inventory policy, by ASIN velocity class).	Required storage. Slow movers are the cube problem; fast movers are the hours problem.
Storage utilization	Cube occupied $\div$ usable cube, weekly.	Above the ceiling: stow rate falls, inbound gets refused, pick paths lengthen. This is the capacity-plan red light.
Inventory health	% of cube held by units with $> N$ weeks of cover; aged inventory; unsellable (damaged, quality hold).	Cube you are paying for that cannot ship. Your S/4HANA story: quality-hold counted as available overstates capacity.

Illustrative storage math. A building with 2.0M usable cubic feet at a 90% ceiling has 1.8M plannable. If its mix averages 0.25 cu ft per unit and the inventory policy is 21 days of cover on 120K daily outbound units, it needs $2.52\text{M units} \times 0.25 = 630\text{K cu ft}$ — 35% utilized, lots of room. Push the same building to 20 days of cover on a 300K/day peak outbound and it needs 1.5M cu ft — 83%, at the edge. The same shelves went from comfortable to critical because *days of cover* and *daily outbound* moved, not the building. That is why capacity planning owns the inventory policy conversation, not just the real-estate one.

7A.4 Outbound — step by step

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Order drop / batching	Customer order is assigned to this building (placement + promise decision). Units are batched into pick work by CPT.	Units dropped per hour by CPT window; work-in-queue hours ahead of each CPT.	Planning logic, not labor. Over-drop and pack drowns; under-drop and pick idles.	No direct cost; it decides every downstream cost. Wrong-building placement adds transportation and transfer cost.
Pick	Associate (or robot brings pod to associate) retrieves unit into a tote.	Units picked/hour (pick rate); picks per tote; pick backlog vs CPT; mispicks per 1,000.	Pick hours; in AR, drive-unit and pod availability; pick-station count.	Variable labor: largest single labor line in most buildings. Rate drops as storage fills (path length) and at peak with new hires on a learning curve.
Rebin / sort (multi-unit orders)	Totes from many pickers are consolidated so one order's units land together.	Units sorted/hour; wall utilization; orders waiting on last unit.	Sort-wall slots; hours.	Variable labor; fixed sort walls / sorters.
Pack	Units go into a box or mailer; box size chosen; dunnage added.	Units or packages packed/hour; packages per hour per station; box-size accuracy (cube out — shipping air is a transportation cost).	Pack stations and hours. Pack is frequently the outbound bottleneck at peak.	Variable labor; variable packaging (corrugate, mailers, tape, dunnage — \$/package, mix-dependent).
SLAM	Scan, Label, Apply, Manifest — package weighed, label printed and applied, manifested to a carrier and a truck.	Packages/hour; weight/dimension mismatches; label errors.	SLAM lines (machines) — a fixed-capacity step.	Fixed equipment; small variable labor and labels.
Ship sort / dock	Package routed to the right outbound door for its carrier, sort center, and CPT; loaded into a trailer.	Packages sorted/hour; CPT attainment (% of planned packages on the truck before it leaves — the headline outbound flow metric); trailer cube fill; late trailer departures.	Sorter capacity; outbound doors; trailer availability.	Fixed sorter; variable load labor; variable transportation — a half-full trailer costs the same as a full one.

Why CPT attainment is the outbound metric that matters. A critical pull time is the moment the promise stops being a plan and becomes a truck. A package that misses its CPT either rolls to the next truck (speed downgrade, possibly a missed promise) or goes on a more expensive mode (premium transportation). Every upstream step — pick, pack, SLAM — is measured against hours-ahead-of-CPT, not against the shift. That is what "flow" means inside a building: work arriving at each step early enough that the last step finishes before the door closes.

7A.5 Downstream — sort center, delivery station, last mile

- **Sort center.** Receives trailers from many FCs, sorts packages by destination delivery station or carrier injection point. Capacity in packages/hour through the sorter and outbound doors by destination. Metrics: sort rate, missort rate, trailer utilization, sort-center CPT attainment to delivery stations.
- **Delivery station.** Receives sorted packages overnight, sorts to routes, dispatches to delivery partners. Capacity in packages/day by route count × packages/route. Metrics: packages per route, dispatch on time, delivery success rate.
- **Why NACF capacity planning cares.** FC outbound capacity is only useful if there is downstream sort and delivery capacity on the same lanes on the same night. A region can have FC slack and still miss speed because the delivery stations near the demand are full. Say "FC capacity and downstream capacity have to be planned on the same page" — it is the regional version of "storage and throughput both."

7A.6 A worked example — demand meets capacity in one region (ILLUSTRATIVE)

Three FCs serve a region. Numbers invented so you can check them.

BUILDING TYPE		USABLE STORAGE (M CU FT, AT CEILING)	OUTBOUND DESIGN RATE (K UNITS/WK)	PLANNABLE OUTBOUND AT 90% (K/WK)	AVG WEEK DEMAND PLACED (K/WK)	PEAK WEEK (×1.6)
FC-A	Sortable AR	1.8	1,400	1,260	1,000	1,600
FC-B	Sortable (manual)	1.2	900	810	700	1,120
FC-C	Non-sort	2.5	500	450	300	480
Region		5.5	2,800	2,520	2,000	3,200

Read it. Average week: 2,000 against 2,520 plannable — 79% utilized, fine. Peak week: 3,200 against 2,520 — a **680K-unit gap**, and it is not evenly distributed: FC-A is 340K over, FC-B 310K over, FC-C is 30K over. "The region is 27% short" is the wrong sentence. "Two sortable buildings are each about 35% short in peak week; non-sort is fine" is the right one.

Check the other currency. FC-A peak inventory: $1.6\text{M/wk} \div 7 = 229\text{K units/day} \times 18 \text{ days of cover} \times 0.25 \text{ cu ft} = 1.03\text{M cu ft}$ against 1.8M usable — 57%, storage is not the problem. FC-B: $160\text{K/day} \times 18 \times 0.30 = 0.86\text{M}$ against 1.2M — 72%, getting tight but not binding. So this is a **throughput** gap in the two sortable buildings, concentrated in pick and pack hours. The lever stack is a labor and flow conversation first, a steel conversation never (for this peak).

Lever stack with illustrative costs for the 650K sortable gap, cheapest and most reversible first:

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
1. Placement shift — move sortable-eligible demand to FC-C's slack and to the next region's sortable buildings	~150K	Extra outbound transportation ~\$0.40/unit on shifted volume (longer zone) → ~\$60K/wk	Fires when week-8 forecast still shows >10% gap. Owner: placement / SCOT partner.
2. Hours — add a shift, extend to 6 days, overtime	~250K (FC-A 150K, FC-B 100K)	Overtime premium 1.5× on ~180K incremental hours-equivalent... illustratively +\$1.2M for the peak period	Labor plan locked at week-10. Owner: FC Ops + regional labor planning.

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
3. Peak hiring with learning-curve haircut	~200K	New hires at ~60% rate for 2 weeks; training and attrition cost; ~\$2M for the period	Requisitions open week-12; ramp complete week-2. Owner: FC Ops / HR.
4. Inbound shaping — pull fast-mover inbound forward to pre-peak weeks, push slow-mover inbound to post-peak	Frees pick/stow hours in peak week (~50K equivalent)	Carrying cost on earlier inventory (small); vendor negotiation effort	Inbound appointment caps by week published week-10. Owner: inbound planning.
5. Overflow site / 3P	Residual	\$1.50–3.00/unit all-in	Only for what 1–4 do not cover. Contract trigger at week-8.
6. Retrofit / automation	+15–30% building throughput	Capex, 2–4 quarters	Not this peak. Goes into the OP cycle for next year.
7. Demand shaping — speed tier, promise	Last resort	Conversion cost (unpriced trust)	Only when a customer-facing miss is the alternative.

What you say at the end of the example. "The plan closes 650K with placement, hours, hiring and inbound shaping, at roughly three to four million dollars of incremental cost for the peak period, and leaves overflow as a priced option with a trigger date. Every lever has an owner and a week it fires. The two things I will not do: spread a pad across all three buildings, which hides that FC-C is fine; and plan either sortable building above 90% of design rate, because congestion arrives before the ceiling does."

7A.7 The labor model — how hours are computed

Required hours (step, week) = units through the step ÷ planned rate (units/hour) for that step

Planned rate = engineered standard × building-specific factor × learning-curve factor × congestion factor

Headcount (step) = required hours ÷ (hours per shift × shifts per week × attendance %)

Total direct hours (week) = Σ over steps (receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve)

Indirect hours = supervision, training, quality, ICQA (inventory counts), safety, maintenance labor

ILLUSTRATIVE FC-A peak week: 1.6M units

pick	$1.6M \div 120/hr = 13,300$ hrs	stow	$1.5M \text{ inbound} \div 110/hr = 13,600$ hrs
pack	$1.6M \div 90/hr = 17,800$ hrs	receive	$1.5M \div 150/hr = 10,000$ hrs
rebin	$0.8M \div 250/hr = 3,200$ hrs	ship/dock	$\approx 4,000$ hrs

direct $\approx 62,000$ hrs; at 36 productive hrs/associate/week and 92% attendance → ~1,870 associates

average week $\approx 1,170$ → peak requires ~700 incremental associates, ~2 weeks of ramp

Two things the interviewer wants to hear you know: **rates are not constants** — they fall when storage is full, when the building is on new hires, and when flow is uneven (idle pack then flooded pack) — and **pack, not pick, is often the peak bottleneck**, because pack rate per unit is lower and pack stations are fixed.

7A.8 The cost model — every dollar a unit picks up

Think of cost per unit as the sum of what the unit touched, divided by how many units shared the building that week. Fixed costs make the denominator matter; variable costs make the process paths matter; congestion makes both worse at the same time.

COST FAMILY	LINES	FIXED / VARIABLE	HOW IT BEHAVES WITH VOLUME AND UTILIZATION
Facility	Lease or depreciation on the building; property tax; insurance; base utilities; security	Fixed	Per-unit cost falls as volume rises — <i>fixed-cost dilution</i> . This is why under-utilized buildings are expensive per unit and why Finance resists new steel until the existing floor is used.
Equipment	Conveyors, sorters, SLAM lines, AR drive units and pods, racking, forklifts, IT/scanners	Fixed (capex, depreciated) + maintenance (semi-variable)	Same dilution. Automation shifts cost from variable labor to fixed equipment — good at high utilization, bad at low.
Direct labor	Receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve	Variable (hours × wage), with step-ups	Roughly linear until the building is full or on new hires; then rate falls and cost per unit rises. Overtime at 1.5×; peak new-hire premium; shift differentials.
Indirect labor	Area managers, process assistants, learning, ICQA, safety, HR, facilities	Semi-fixed (steps with shifts and headcount bands)	Moves in steps when you add a shift; otherwise flat.
Packaging	Corrugate, mailers, tape, dunnage, labels	Variable per package	Mix-driven. Box-size accuracy also drives outbound transportation (dimensional weight).
Inbound transportation	Linehaul vendor → IXD/FC; IXD → FC transfers; drayage	Variable per trailer / per unit	Driven by placement: a unit stowed far from its vendor and far from its demand pays twice. Trailer cube fill is the efficiency metric.
Outbound transportation	FC → sort center linehaul; sort center → delivery station; last mile; carrier injection	Variable per package, zone and weight dependent	Usually the largest cost family after labor. Driven by placement (distance to customer), speed tier (air vs ground), package cube, and trailer fill. A placement shift to close a capacity gap is paid here.
Transfers and rebalancing	FC ↔ FC moves to fix placement after the fact	Variable	A tax on bad placement or bad forecasting. Track it as its own line so it cannot hide in inbound.
Inventory carrying	Cost of capital on units on hand; shrink; damage; obsolescence/markdown	Variable with inventory level	Rises with days of cover. Inbound shaping (pull-forward) spends this to save labor in peak week — price the trade.
Quality and defects	Mispicks, mis-ships, damages, concessions, returns processing	Variable per defect	Rises when rates are pushed past standard. A building running hot pays here two weeks later.
Overflow / 3P	Short-term space and labor contracts	Variable (all-in per unit) with minimums	Expensive per unit but no fixed commitment — the right lever for a short peak, the wrong one for a structural gap.

Cost per unit, illustrated at three utilizations (same building, one week, invented numbers):

	60% utilized	85% utilized	97% utilized ("full")	
Units shipped	840K	1,190K	1,360K	
Fixed (facility+equipment)	\$1.70M → \$2.02/u	\$1.70M → \$1.43/u	\$1.70M → \$1.25/u	
Direct labor	\$2.0M → \$2.38/u	\$2.7M → \$2.27/u	\$3.6M → \$2.65/u	← rates
fall, OT rises				
Indirect labor	\$0.45M → \$0.54/u	\$0.50M → \$0.42/u	\$0.55M → \$0.40/u	
Packaging	\$0.55/u	\$0.55/u	\$0.55/u	
Outbound transport	\$3.20/u	\$3.20/u	\$3.45/u	← CPT
misses → premium mode				
Defects / concessions	\$0.06/u	\$0.08/u	\$0.18/u	
Total	≈ \$8.75/u	≈ \$7.95/u	≈ \$8.48/u	

Read it. Cost per unit is U-shaped in utilization. Too empty and fixed cost is undiluted; too full and labor, transport and defects spike. The economic optimum sits near the operating ceiling, not at 100%. That is the whole argument for planning to a ceiling, and it is the argument you make to Finance in their own currency: "the last 7% of utilization costs more per unit than the first 85% saved."

Inbound vs outbound cost, said plainly. Inbound cost per unit (freight + receive + stow) is small per unit but front-loaded and placement-sensitive. Outbound cost per unit (pick + pack + packaging + transportation) is larger and speed-sensitive. Placement is the one decision that moves both: put the unit near its demand and you pay more inbound and less outbound; the capacity plan has to know which side of that trade a region is on.

7A.9 The WBR this role would run

Outputs are what the business feels. Inputs are levers a named owner can move this week. Every line has a grain, a target, an owner, and a red that arrives pre-decomposed: which input moved, whose lever, what correction, verified by the same number next week. Illustrative targets.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Outputs				
O1	Capacity delivered vs plan (units/week the network could process)	Region × building type × week	≥ 100% of committed plan	You. The headline: did the capacity exist when the plan said it would.
O2	Throughput utilization vs ceiling	Building × week (peak: × day)	75–90% band	You with FC Ops. Below band = cost; above = speed and defects.
O3	Storage utilization vs ceiling	Building × week	≤ 90%	You with inventory planning. The second currency.
O4	Customer speed attainment (promised speed met)	Region × speed tier × week	≥ target by tier	Shared with S&OP / transportation. The customer-facing reason the plan exists.
O5	Cost per unit vs plan (fully loaded, and by family)	Building × week; region × month	≤ plan	Finance partner. The second half of the charter.
O6	CPT attainment	Building × CPT × day	≥ 98%	FC Ops. Outbound flow health; leading indicator of O4.
O7	Dock-to-stow hours	Building × day	≤ 24 h	FC Ops / inbound. Inbound flow health; leading indicator of refused appointments.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Inputs				
I1	Forecast accuracy at planning grain (and bias)	Building type × region × week, 4- and 8-week horizons	MAPE within band; bias ≈ 0	S&OP / SCOT. A biased forecast is a capacity error wearing a demand costume.
I2	Placement compliance (units landed in the planned building)	Region × week	≥ 95%	Placement partner. Off-plan placement is the silent capacity thief.
I3	Labor plan attainment (hours staffed ÷ hours required, by step)	Building × step × week	≥ 97%	FC Ops / labor planning.
I4	Rate attainment vs planned rate, by step	Building × step × week	≥ 95% of plan	FC Ops. Where congestion and learning curve show up first.
I5	Launch and retrofit milestones on dated gates	Project × gate	Green/red vs date	Launch / Engineering / Real Estate. New capacity that is late is a gap you planned on.
I6	Inbound appointment adherence and vendor compliance defects	Building × week	≥ 95%; defects ≤ x per 1,000	Inbound planning / vendor management.
I7	Inventory health: days of cover by velocity class; % cube in aged / unsellable	Building × week	Within policy; aged ≤ x%	Inventory planning. Cube you cannot sell is capacity you do not have.
I8	Trailer cube fill, inbound and outbound	Lane × week	≥ target	Transportation. The cheapest transportation dollar.
I9	Transfers and rebalancing volume and cost	Region × week	Trending down	You. The tax on bad placement, kept visible.
I10	Lever status: which planned levers have fired, on which trigger, at what cost	Lever × week	All triggered levers fired on time	You. The plan is a list of triggers; this line proves it is alive.

Page one vs annex. Page one: O1–O7, I1–I10, RAG with week-over-week deltas, five callouts, and the action log with owner and date. Annex: forecast histograms, rate curves by building, placement heatmaps, cost bridges, launch Gantts. **Correction of error** is written when O1 misses plan by more than a published threshold, when O2 or O3 exceeds ceiling for two consecutive weeks, or when a lever that should have fired did not. It ends in a mechanism change — a trigger moved earlier, a ceiling re-measured, an owner named — never in "try harder."

7A.10 How you would model it

Four layers, each feeding the next. Build the simplest version of each first; add fidelity where the plan is actually sensitive.

LAYER 1 – DEMAND	units by week × region × building type × speed tier, from S&OP peak factors by week; a range (P10 / P50 / P90), not a point – the plan is sized to a quantile you choose on purpose
LAYER 2 – PLACEMENT	which building absorbs which demand: minimize inbound + outbound transport + transfer cost, subject to each building's storage and throughput ceilings → units/week per building. The network LP lives here; SCOT owns the engine, you own the constraints and whether the answer is plannable
LAYER 3 – BUILDING	per building per week: required hours by step (7A.7) and required cube (7A.3) vs available; gap by currency by step; rate curves as a function of utilization and new-hire share; CPT structure (how many trucks, when) – flow inside the week, not just the weekly total
LAYER 4 – COST	per building per week: fixed + labor + packaging + transport + carrying + defects (7A.8); cost per unit; incremental cost of each lever; the U-curve
OUTPUTS	gap by week × building × currency; ranked lever stack with cost and lead time; triggers; sensitivity – which forecast error, rate miss or launch slip breaks the plan
VALIDATION	backtest on last year's weeks: did the model's "available" match what the buildings shipped? Where it missed, which layer was wrong – demand, placement, rate, or ceiling?

Three modeling judgments to say out loud. (1) Plan to a quantile, not a point: if you size to P50 you will be short half the time. Decide with Finance which quantile the cost of shortage justifies. (2) Rates are curves, not constants: a model with fixed rates will tell you a 97%-utilized building is fine. It is not. (3) Weekly totals hide CPT misses: a building can hit its weekly number and miss every 6 PM truck. Model at least the daily CPT structure for peak weeks.

What you would ask on day one. What is the current ceiling by building type, who set it, and when was it last re-measured against actuals? Does placement see the throughput ceiling or only storage? Which lever fired last peak that was not in the plan? Where does forecast bias live by region? And: which of these numbers does Finance accept, and which does it dispute?

7A.11 Whiteboard math — the ten computations you must be able to do cold

Each one: the formula, why it is the right formula, a worked number (ILLUSTRATIVE), and what it tells a capacity planner. Do them on paper once before Friday. In the room, say the formula first, then the numbers, then the implication.

1. Peak sizing from a forecast with error — plan to a quantile, not a point

Required capacity at service level α = Forecast_P50 × (1 + z_{α} × CV)
 CV = forecast error ($\sigma \div$ mean) at the planning grain and horizon
 $z_{0.80} \approx 0.84$ $z_{0.90} \approx 1.28$ $z_{0.95} \approx 1.65$

Worked: peak-week P50 = 1.6M units at FC-A; 8-week-ahead CV = 12%
 $P80 = 1.6M \times (1 + 0.84 \times 0.12) = 1.76M$
 $P90 = 1.6M \times (1 + 1.28 \times 0.12) = 1.85M$
 $P95 = 1.6M \times (1 + 1.65 \times 0.12) = 1.92M$

Implication. Sizing to P50 is short half the time. The quantile is a *business decision*: the cost of being short (premium transport, speed misses, overflow at spot rates) vs the cost of being long (undiluted fixed cost, idle hours). Choose it with Finance, write it down, and size every lever to it. If CV is 12% at 8 weeks and 6% at 2

weeks, the plan can hold reversible levers (hours, placement) at P90 and commit irreversible ones (overflow contracts) at P80, then upgrade as the forecast tightens.

2. Storage required and utilization — the cube currency

```
Inventory units    = DailyOutbound × DaysOfCover
Cube needed       = Inventory units × cube_per_unit(mix) ÷ BinFillEfficiency
Utilization       = Cube needed ÷ (GrossCube × Ceiling)
```

Worked (FC-A, same as 7A.6):

Inventory = 229K units/day × 18 days = 4.12M units; × 0.25 cu ft = 1.03M cu ft of product

Usable = 2.86M gross bin cu ft × 0.70 bin fill × 0.90 ceiling = 1.80M cu ft

Utilization = 1.03 ÷ 1.80 = 57%

Sensitivity: +1 day of cover = 229K × 0.25 = 57K cu ft ≈ 3.2 points; +10 days ≈ 32 points → 89%, at the ceiling

Implication. Days of cover is the biggest storage lever and it is an inventory-policy decision, not a real-estate one. Three days of cover is ten points of utilization — the difference between comfortable and congested. The capacity planner must own the days-of-cover conversation with inventory planning.

3. Throughput required vs available — the hours currency

```
Required hours_s  = Units_s ÷ Rate_s
Available hours_s = Heads_s × ShiftHrs × Shifts/wk × Attendance
Gap_s             = Required - Available ; binding step = largest positive gap
```

Worked (FC-A peak week, 1.6M outbound):

pack: 1.6M ÷ 90/hr = 17,800 hrs needed; staffed 420 heads × 8 h × 5 days × 0.92 = 15,460 hrs

gap = 2,340 hrs ≈ 210K units at pack → pack binds even though pick (13,300 hrs needed) is staffed to 14,000

Fixes priced in hours:

+1 shift-day/wk on pack = 420 × 8 × 0.92 = +3,090 hrs at 1.0× wage

OT on existing heads = 2,340 hrs at 1.5× wage

move 210K units to FC-C = transport Δ (7A.6: ~\$0.40/unit)

Implication. A building's throughput is its slowest step *at the hour the truck leaves*. Always compute the gap per step; the binding step is where the money goes. Pack binds more often than pick because its rate is lower and stations are fixed.

4. Little's Law — turning a flow metric into a backlog and back

```
WIP = Throughput × CycleTime      ⇔      CycleTime = WIP ÷ Throughput
```

Inbound: 215K units/day arriving, dock-to-stow target 24 h

→ WIP allowed on the inbound floor = 215K × 1.0 day = 215K units

if the floor holds 430K units, dock-to-stow is running 48 h, whatever the dashboard says

Outbound: a CPT with 180K units due in 6 h needs 30K units/h through pack

at 90 units/h that is 333 pack heads working that CPT

Implication. You can measure cycle time by counting what is on the floor. Dock-to-stow hours, backlog units and receive rate are one relationship, not three metrics; if two are green and one is red, one of the greens is wrong.

5. Congestion — why cost per unit turns up past the ceiling

Queueing approximation: $Wait \propto \rho \div (1 - \rho)$, ρ = utilization
 $\rho = 0.70 \rightarrow 2.3$ $0.85 \rightarrow 5.7$ $0.90 \rightarrow 9.0$ $0.95 \rightarrow 19$ $0.97 \rightarrow 32$ (relative wait)

Storage analog – stow rate falls as empty bins get scarce (illustrative curve):

$Rate(\rho) = DesignRate \times [1 - k \times \max(0, \rho - 0.85)^2 \div 0.0225]$, $k \approx 0.3$

$\rho = 0.85 \rightarrow 100\%$ of design $0.90 \rightarrow 97\%$ $0.95 \rightarrow 87\%$ $1.00 \rightarrow 70\%$

Implication. Going from 90% to 95% utilization doubles waiting; going to 97% triples it again. That is the arithmetic behind "plan to a ceiling, not to 100%." It is also why the last five points of utilization cost more per unit than they save in fixed-cost dilution (7A.8).

6. Fixed-cost dilution vs variable-cost inflation — the U-curve, derived

$CPU(p) = \text{Fixed} \div \text{Units}(p) + \text{VarLabor}(p) + \text{Packaging} + \text{Transport}(p) + \text{Defects}(p)$
 Fixed $\div$ Units falls like $1/p$
 VarLabor, Defects rise past the ceiling (Rate(p) falls, queueing grows)
 Transport rises when CPT misses force premium modes

Worked (7A.8 numbers): CPU at 60% = \$8.75 at 85% = \$7.95 at 97% = \$8.48
 60 $\rightarrow$ 85 dilution: fixed/unit \$2.02 $\rightarrow$ \$1.43 (-\$0.59); labor/unit \$2.38 $\rightarrow$ \$2.27 (-\$0.11)
 85 $\rightarrow$ 97 inflation: fixed/unit \$1.43 $\rightarrow$ \$1.25 (-\$0.18); labor +\$0.38, transport +\$0.25, defects +\$0.10 (+\$0.73 total) $\rightarrow$ net +\$0.53 per unit

Implication. The minimum of the curve is the economic ceiling, and it is *below* the physical ceiling. When Finance asks why you will not plan to 100%, this is the table you show: the last 12 points cost 53 cents a unit.

7. Lever cost per unit of capacity — ranking the stack

Lever cost (\$ per unit of gap closed) = Incremental cost $\div$ Units of gap it closes
 Rank ascending, subject to lead time $\leq$ weeks remaining to peak

Worked (7A.6 lever stack, 650K/wk gap over a 4-week peak):

placement shift	\$60K/wk $\times$ 4 $\div$ (150K $\times$ 4) $\approx$ \$0.40/unit	lead 1-2 wks	reversible
inbound shaping	carrying cost on pulled-forward stock $\approx$ \$0.30/unit	lead 6-8 wks	reversible
OT / extra shift	\$1.2M $\div$ (250K $\times$ 4) $\approx$ \$1.20/unit	lead 4-6 wks	reversible
peak hiring	\$2.0M $\div$ (200K $\times$ 4) $\approx$ \$2.50/unit	lead 10-12 wks	semi-reversible
overflow / 3P	\$1.50-3.00/unit all-in	lead ~8 wks	contractual
demand shaping	conversion loss — unpriced, so last		

Implication. Flow before hours, hours before hiring, hiring before overflow, overflow before the promise — but the ranking is an output of this arithmetic, not a slogan. If placement shift costs \$0.40 and the next region has no slack, it drops off the list and hours move up.

8. Build vs no-build — the break-even forecast

Build (annualized) = Capex $\div$ Life + Lease/fixed opex + Fixed labor $\leftarrow$ paid regardless of volume
 No-build (annual) = $\sum \text{weeks} [\text{Overflow\$} + \text{Transfers\$} + \text{PremiumTransport\$} + \text{SpeedDowngrade} \times \text{ConversionCost} + P(\text{peak miss}) \times \text{PeakMiss\$}]$
 Break-even demand D^* = the demand at which the two are equal; report it as a forecast quantile

Worked (illustrative): build = \$28M/yr all-in
 no-build cost at P50 demand = \$19M/yr; at P75 = \$31M/yr; at P90 = \$44M/yr $\rightarrow D^* \approx P68$
 sizing to P80: the build pays. sizing to P50: it does not, and overflow is the plan.

Implication. Do not argue "we need the building." Show D^* , show where the planning quantile sits, and let the decision fall out. Also show the middle path — retrofit for +20% throughput at a third of the capex — on the same axis.

9. Learning curve and attrition — why peak heads are not peak hours

Effective hours (wk) = $\sum_{\text{cohorts}} \text{Heads}_c \times \text{ShiftHrs} \times \text{Attendance} \times \text{Productivity}_c(\text{weeks since hire})$
Productivity $\approx$ 0.60 (wk 1) 0.80 (wk 2) 0.95 (wk 3) 1.00 (wk 4+) [illustrative]
Ramp attrition $\approx$ 10–15% of a cohort $\rightarrow$ hire $\sim 1.15\times$ the heads you need

Worked: need +700 effective associates in peak week 0
hire 300 in wk -3 and 500 in wk -2 (800 gross)
effective in wk 0 $\approx 300 \times 0.95 + 500 \times 0.80 = 685 \rightarrow$ 15 short, borderline $\rightarrow$ add 50 to the wk -3 cohort

Implication. A labor plan in heads is wrong by the learning curve. Plan in effective hours by cohort, and the hiring trigger moves earlier than the heads math suggests.

10. Speed — the CPT cliff and what a late unit costs

A unit that misses its CPT by 1 minute loses a full truck cycle — often a calendar day of promise.
Speed loss (units/day) = $\sum_{\text{CPT}} \max(0, \text{Units due at CPT} - \text{Pack capacity in the window before CPT})$
Cost per missed unit $\approx$ premium-transport uplift (if upgraded)
or $P(\text{cancel} \mid \text{late}) \times \text{margin} + \text{trust (unpriced)}$

Worked: 18:00 CPT has 180K due; pack capacity 14:00–18:00 = 4 h $\times$ 36K/h = 144K $\rightarrow$ 36K miss, daily
60% upgraded to premium at +\$2.50 $\rightarrow 21.6K \times \$2.50 = \$54K/\text{day}$ in transport alone, before cancels

Implication. Weekly utilization can be 80% and the 18:00 CPT can miss every day. Capacity must be planned at the CPT window grain for peak weeks. This is Case C in Section 7, in numbers.

7A.12 Lines for this section

- "Capacity is two currencies — cube and hours — and the plan is honest only if both are checked at the step where each binds."
- "Dock-to-stow is inbound flow. CPT attainment is outbound flow. Everything else is a component."
- "Cost per unit is U-shaped in utilization. The optimum is near the ceiling, not at a hundred percent."
- "Placement is the one decision that moves inbound and outbound cost at the same time."
- "A biased forecast is a capacity error wearing a demand costume."
- "Inbound shaping spends carrying cost to buy peak-week labor. Price the trade; don't just do it."
- "Pack, not pick, is usually where peak breaks."

7. The functional block — capacity planning

You will get at least one of these. Format they expect: clarify out loud (60–90 seconds) $\rightarrow$ structure (60 seconds) $\rightarrow$ a math checkpoint $\rightarrow$ recommendation with owners, triggers and risks. Say "both constraints — storage and throughput" early in every answer. Label every number you make up as illustrative, and say what you would replace it with on day one.

Primer: the shape of a capacity plan

Demand side	units by week × building type × speed tier, with a peak factor (forecast owner: S&OP / SCOT; your job is to interrogate its grain and mix)
Required capacity needed	storage: units on hand ≈ daily outbound units × days of cover → cube throughput: units/week each process step must push (inbound, stow, pick, pack, ship) both, by building type; the binding one is the one you plan to
Available capacity before 100%)	buildings × design rate × a utilization ceiling (congestion starts before 100%) minus planned downtime, retrofits, launches still ramping
Gap by week	required – available, by region, by type; where and when it binds
Leverage, cheapest and fastest first	
1. flow: move units – placement / allocation to buildings with slack (days to weeks) 2. hours: shifts, overtime, labor plan (weeks) 3. inventory policy: pull inbound forward or push it back; trim slow movers (weeks) 4. overflow: 3P / temporary space (weeks to months) 5. retrofit / automation: more throughput in the same cube (quarters) 6. expansion / new build: more cube (OP cycle; years) 7. demand shaping: speed tier, promise, selection (the lever nobody wants to admit)	
Plan	named owner per lever, trigger that fires it, review cadence, and what you will not do

Case A — "Build next year's capacity plan for a region."

Clarify (say these out loud). Which building types are in scope? Is the demand forecast handed to me or do I own it? What is the peak factor and what week does it peak? What are the policy ceilings on utilization? Which launches and retrofits are already committed? What is the planning grain — week, building, type?

Structure. The primer, top to bottom. Spend the most time on the gap-by-week and the lever ranking.

Math checkpoint (illustrative). "If a building ships 1.0M units a week at average and the peak factor is 1.6, peak week is 1.6M. If its throughput design rate is 1.4M and the ceiling I am allowed to plan to is 90%, usable is 1.26M. The gap is 340K units in peak week. Flow first: how much of that can neighboring buildings absorb inside their own ceilings? Then hours. Overflow only for what is left. I would replace every one of those numbers with the S&OP demand file and the engineering design rates on day one."

Recommendation. A one-page plan: gap by week by region, the lever stack with owners, the triggers that promote a lever (e.g., "if the week-8 forecast is still 5% above plan, overflow contracts sign"), and the two things you will not do — plan to 100% of design rate, or pad every building's forecast to hide a placement problem.

Risk: forecast mix shifts between types; you plan by type, not by total.

CASE A — THE PROBES THEY WILL RUN INSIDE THE CASE, AND THE ANSWERS

"Where does your demand number come from, and how much do you trust it?" From S&OP/SCOT at week × region × building type × speed tier. I trust it as a distribution, not a point: I ask for the error (CV) at the 8-week and 2-week horizons and the bias by region, and I size reversible levers at P90 and irreversible ones at P80 until the forecast tightens (7A.11 #1).

"Which constraint binds first in your example — storage or throughput?" Throughput, in the two sortable buildings, specifically pack hours before the evening CPTs. Storage is at 57% and 72%. I know that because I ran equations (2) and (4) separately per building; the network total would have hidden it.

"Why 90% and not 95%?" Queueing: waiting roughly doubles between 90 and 95 (7A.11 #5), stow and pick rates fall as bins fill, and the cost-per-unit curve turns up past the economic ceiling (7A.11 #6). If the building's own measured rate curve says the knee is at 93%, I plan to 93%. The 90 is a placeholder for a measured number.

"Your placement shift costs transport. How much, and who pays?" Illustratively \$0.40 a unit on shifted volume — longer zone. Transportation's budget pays; the capacity plan's cost line shows it. I would not hide a capacity fix inside someone else's cost center; the WBR line I9 (transfers and rebalancing) exists so this stays visible.

"What if the forecast is wrong by 15% the other way — you over-hired?" The levers are sequenced by reversibility for exactly this: placement and OT unwind in a week; hiring is the first semi-irreversible lever and it fires at week -12 only if the P80 still shows the gap. If demand comes in low, the cost is undiluted fixed cost and some idle hours — I would show that number next to the cost of being short, and that comparison is how we chose the quantile in the first place.

"How do you know the lever actually fired and worked?" WBR line I10: each lever has a trigger date, an owner, and a 'fired / not fired / effect measured' state. Effect is measured in the same currency as the gap — units of pack capacity in the week it was needed — not in 'we added a shift.'

"What would you not do?" Spread a pad across all three buildings — it hides that FC-C is fine and spends everywhere. Plan either sortable building above its ceiling. Pull the promise before the priced levers are exhausted. And I would not present a plan without the quantile written on it.

"How would you present this to Finance?" Two columns, same rows (S7): cost of the plan by lever versus cost of the shortfall it prevents — overflow at spot, premium transport, speed downgrades at conversion cost, risk-adjusted peak miss. Break-even demand marked on the forecast distribution (7A.11 #8). Six pages, the table on page one.

"Walk me through the data pull." Demand: forecast file at week × building type × region × speed tier with error by horizon. Capacity: design rates by process path by building from Ops Engineering; measured rates by week for the last 52 weeks to fit the utilization curve; usable cube and bin fill by building. Labor: heads, attendance, cohort start dates. CPT structure: trucks and due units by hour. Costs: wage rates, OT premium, lease and depreciation by building, transport \$/unit by lane, overflow quotes. Join everything at building × week; keep CPT × day for peak weeks.

If they ask for it as a STAR ("tell me about a time you balanced capacity and demand"): S2 is the lead — demand exceeded the apparent capacity of the network at 4 AM; the task was 99% fill; the actions were promise-at-risk ranking and rebalancing to nodes with hours; the result was 99% held and \$8M protected — and the capacity turned out to be a planning-system failure, which is the lesson. S1 is the backup: consolidation as a capacity decision, throttled against a quality gate.

Case B — "Peak forecast just came in 10% above capacity. Go."

1. **Verify the number before you spend on it.** Grain and mix: is the 10% real demand, or a type-mix shift that makes one building type bind while another has slack? Same cohort weights as last forecast?
2. **Find where it binds.** Storage or throughput? Inbound dock, stow, pick, pack, or outbound? One region or network? "A network 10% is usually three buildings at 25% and the rest fine."
3. **Cheapest lever first, with a trigger.** Placement and allocation shifts this week. Labor plan next. Pull slow-moving inbound back; pull fast-moving inbound forward. Overflow for the residual. Speed tier only if the customer promise is the last protection.
4. **Publish the plan with a date on every lever** and a freeze on non-essential change into peak — the UDC October-to-December freeze.
5. **What you will not do.** Spread a national pad across every building. That hides the cell that is actually breaking and spends cost everywhere.

Story to attach: S2 — rank by promise-at-risk, rebalance across nodes, timestamped log, then a mechanism so you never discover the regime in November again.

CASE B — PROBES AND ANSWERS

"You have one hour. What's the first number you pull?" The gap by building by step, not the network 10%. Equation (4) per building: units due at each CPT vs pack-and-ship hours in the window. Then (2): any building above 85% storage. Ten percent on the network is usually three buildings at 25–35% and the rest fine (7A.6).

"Is it real demand?" Three checks before spending: same cohort weights and promotions as last forecast; mix shift between building types (a 10% rise in non-sort with sortable flat is a different plan); and bias — if this region's 8-week forecast has run 6% high for a year, the 10% is 4%.

"What fires this week?" Placement to buildings with slack, with the transport delta priced (19). OT authorization on the binding step in the binding buildings. Inbound appointment caps on slow movers into those buildings, pulling fast movers forward. All reversible.

"What fires only on a trigger?" Hiring at week –12 if P80 still shows the gap. Overflow contract at week –8 for the residual above what hours and placement close. The promise lever only when the priced levers are exhausted — and I would write the dollar value of a one-day speed downgrade next to it so nobody pulls it casually.

"How do you communicate it?" One page by Friday: gap by building by week by currency, the lever stack with cost per unit closed, triggers with dates, owners, what I will not do, and the number I will report next week to show whether the first levers worked.

Case C — "Utilization looks fine but speed is missing."

Utilization is a weekly average. Speed misses live at the hour, the dock door, the sort type, the cutoff. Change the grain: by building by hour by process step. Check mix: did fast-promise demand land in buildings far from the customer? Check cutoff cliffs: units arriving an hour late lose a calendar day. **Story:** S5 — the reports were wrong in both directions because they measured the wrong grain. **Line to say:** "If I am still talking about network utilization after two follow-ups, I am failing the question."

CASE C — PROBES AND ANSWERS

"Give me the math." 7A.11 #10: a CPT with 180K due and 144K of pack capacity in its window misses 36K units every day while weekly utilization reads 80%. Speed is lost at the CPT window, not the week.

"What else could it be?" Placement — fast-promise demand landing in buildings far from the customer, so the unit ships on time and still arrives late; downstream — sort center or delivery station capacity on that lane; or a dock-to-stow backlog making units unavailable to promise at order time (Little's Law, 7A.11 #4).

"Which one first?" CPT attainment by building by CPT by day (O6) splits it: if CPT attainment is high and speed is low, the problem is placement or downstream; if CPT attainment is low, it is inside the building, and the step with the largest hours gap in the window is the cause.

Case D — "Finance says the new building is not justified."

Do not argue. Price both columns. Cost of the build, annualized. Cost of not having it: overflow contracts, transfers and split shipments, speed downgrades and their conversion cost, labor premiums, the risk-adjusted cost of a peak miss. Show the sensitivity: at what forecast does the build break even? Offer the middle path — retrofit or flow redesign — priced in the same table. **Story:** S7 — the two-column table that flipped an eighteen-month plan; Finance attacked it and it survived. **Line:** "I would rather lose this argument to a better number than win it with a louder voice."

CASE D — PROBES AND ANSWERS

"What's in the no-build column?" Overflow at contract or spot rates for the shortfall weeks; transfers and rebalancing; premium transportation for CPT misses; speed downgrade at conversion cost; labor premium from

running existing buildings past the economic ceiling (7A.11 #6); and a risk-adjusted cost of a peak miss — probability × cost — which I state as an assumption and let Finance change.

"Where's the break-even?" 7A.11 #8: the demand level D^* where annualized build cost equals annual no-build cost, reported as a quantile of the forecast — "the build pays at P68 and above." Then the question is not "is the building justified" but "what quantile are we planning to" — a decision Finance owns with us.

"What if they still say no?" Commit. Then the no-build plan becomes the plan: overflow contracts signed by week -8, the transport and labor premiums budgeted, and the speed exposure written down. I bring the realized no-build cost back to the OP cycle next year as data, not as an argument.

Case E — "Define the metrics for a network capacity program."

Outputs (what the business feels): capacity delivered vs plan by week; utilization vs ceiling by type; cost per unit; customer speed attainment. Inputs (what a named owner can move this week): launch and retrofit milestones on dated gates; placement compliance; forecast accuracy at the planning grain; labor plan attainment; inbound ASN accuracy. Mechanism: a weekly review where reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week — and a correction-of-error write-up for any miss above a published threshold that ends in a mechanism change, not "try harder." **Line:** "Inputs without outputs are a hobby. Outputs without inputs are weather."

CASE E — PROBES AND ANSWERS

"Pick the one output you'd be judged on." Capacity delivered vs plan (O1), because it is the promise this team makes to everyone else. Cost per unit (O5) is the constraint on how I deliver it.

"Pick the one input you'd watch hardest." Forecast bias by region (I1), because a biased forecast makes every other metric lie in the same direction, and placement compliance (I2) a close second, because off-plan placement is the silent thief.

"How do you stop the WBR from becoming a status meeting?" Reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week. Definitions frozen (S8). A correction of error above a published threshold that ends in a mechanism change. And a line (I10) that shows whether triggered levers actually fired.

"What's the full list?" 7A.9: seven outputs, ten inputs, each with grain, target and owner.

Fermi, if they throw one

- State assumptions. Compute out loud. Label it illustrative. Say what you would replace it with.
- *"How many units does a large sortable FC ship in a week?"* Structure: shifts × hours × headcount × units per hour, cross-checked against outbound dock capacity. Do not quote an Amazon number you have not seen. "I would take the design rate from Ops Engineering on day one."
- *"How much storage for a week of demand?"* Daily outbound × days of cover ÷ units per cubic foot ÷ bin utilization. Say the structure, pick round numbers, flag the two you are least sure of.
- *"Cost of a 4-point speed miss at peak?"* Volume × delta × cost per late unit, plus the unpriced trust cost. Name the grain. Never say "huge."

Lines that read as judgment, not worship

- "I plan to a ceiling, not to design rate. Congestion starts before the building is full."
- "A network average hides the three buildings that are actually breaking."
- "Flow is the cheapest capacity in the building. Steel is the most expensive. I spend them in that order."
- "The forecast is an input I interrogate, not a fact I plan to."
- "Every lever gets a trigger and an owner before peak, or it is a wish."

8. The hard questions

"You have never built an Amazon FC capacity plan. Why you?"

"Correct, and I will not pretend otherwise. An internal transfer brings Amazon scar tissue. I bring the three things a capacity plan is made of, each proven: allocating demand across a network when the plan did not fit — 99% fill through peak; consolidating buildings and running a launch behind gates — plus fifteen percent, zero incidents; and turning measured data into planning parameters — two hundred lanes, a million and a half released. I also bring the review grammar this team already runs — input levers, output metrics, pre-wired reds. Days one to thirty I learn your numbers, your building types, and the five decisions last peak that nobody wants to talk about. I will not arrive trying to rebuild it. I will arrive with the mechanism."

"This role is in Bellevue."

Decide before Friday and say it in one sentence. Do not improvise a relocation answer in the room. [Your line here.]

"Tell me about a failure."

Pick one, own it in the first sentence, name the mechanism it built, stop. Three ready:

- **UAT that crashed under load (Elementum).** Five users, ten scenarios; I sent screenshots instead of digging. Concurrent load crashed it; sign-off missed; sponsor escalation. The implicit requirement — performance at the customer's shape — was not in the entrance gate. Every critical UAT since has numeric load criteria; Flexport's launch proof was 500 containers in 20 minutes at 99.6% connectivity before a customer touched it. "I was the mechanism that failed; I wrote the mechanism that replaced me."
- **RF devices, 1.5-week training slip (Sephora JDA).** In S9. Single-point dependency in front of an exit criterion now gets a what-if plan regardless of size.
- **Security review, two weeks (Flexport Snowflake).** In S8. "Whose approval does this data need, and what is its lead time" is now a kickoff question.

"Tell me about a time you had to influence without authority, two levels up."

S7. Then one sentence: "I set the instrument, not the person — a named line, a numeric target, a published grain, a date. Next Monday's number is the appeal."

"How do you set goals for people who don't report to you?"

"I set the instrument, not the person. A named input line, a numeric target, a published grain, and a date the number gets checked. On UDC the issues log made it unemotional: an owner's row either moved or it did not. I have never won a goal argument with a title; I have won them with a line on the page everyone reads."

"What would you do in your first 90 days?"

"Thirty days: learn the plan as it actually exists — the demand file, the design rates, the ceilings, which lever fired last peak and why. Sixty: take over the weekly review's output block and pre-wire the reds. Ninety: one output metric moved by a mechanism the review can audit — a trigger that fires a lever before the week it is needed, or a placement change that bought capacity without steel. Not a new dashboard."

"Are you an L6 or an L7?"

If asked, do not oversell. "L6 is running the plan and the review well, independently, on complex problems with real trade-offs. That is what I have done. I have also flipped a socialized roadmap with a table, published

tolerances that stopped go-lives, and throttled my own launch. I have not held a large direct-report org, and I will not pretend that is my differentiator."

9. Your questions — pick two

1. "When the capacity plan was wrong last peak, how long did it take to know which input was wrong — the forecast, the design rate, or the placement?" (*Shows you think in decomposition. Whatever they name, you have the matching story.*)
2. "Where does the storage-versus-throughput call actually get made today — in the model, in a planning table, or in a negotiation between S&OP and Ops?" (*The answer is almost always "it's complicated," which is your opening.*)
3. "What does PPT own relative to this team, and where do the two of you most often disagree?"
4. "Which lever is the team least willing to pull — flow, hours, overflow, or the promise — and why?"
5. "What does a good first ninety days look like to you, in one output metric?"

10. Close — fifteen seconds

"I want this work. I have not built an Amazon capacity plan, and I have said so. I have allocated a network through peak, consolidated buildings behind gates, and turned measured data into the parameters a plan runs on. In ninety days I will move one output with a mechanism your review can audit. If that is the job, I am here for it."

11. Numbers worth memorizing

All on your resume or in your war book. If you cannot say what sits under one, leave it out.

NUMBER	WHAT IT IS
3 mega-DCs, ~100 stores, 99% fill, ~\$8M, ~20% stockouts	Manhattan allocation and promising through peak; Cyber Monday 4 AM
200 lanes, p99, 99% LT accuracy, 580K units, ~\$1.5M	Elementum lead-time distributions replacing brochure numbers
70% → 85%, ~\$2M, zero recon drift	Elementum forecast accuracy and safety stock; MD04/MB52 daily recon
40% / 55% / 99%, ~\$8M, 25 locations	Austria: reported vs true data quality; freight savings
~1,000 routes, ≥95% valid ASN, 60% → 95%	EDI onboarding gate; shipment completeness
7 → 68, \$0.66 → \$0.03, 72% → 90% at 5x, 2 months early	Flexport visibility platform; six graduation criteria
~50% vs ~25%, 66%, \$5M, 6 vs 18 months	Roadmap flip on cost-to-serve
4 → 1, 60% → 90%, >\$100K/yr, 1% parallel run, 4% first catch	Snowflake WBR pipeline
8 months, 7% → 1.8% → 0.3% vs 0.5%, ~3% under \$1.8M, \$20M	Dual-DC JDA migration; mocks; pipeline unblocked
15+ tracks, 60+ stakeholders, 836 rows, Nov 22 / Dec 20 / Jan 30	UDC omni DC program; test gates; Oct–Dec freeze

NUMBER	WHAT IT IS
343 tasks; sync < 99% × 4 h or p95 > 2× × 1 h; never fired; MCS off; 0 incidents	S/4HANA ↔ ISM cutover runbook and rollback triggers; legacy inventory system retired
Returns plant; RTV STOs as pick-reserve; receive → grade → restock / scrap / RTV	S/4HANA stock-state model: nothing promiseable until disposition; credit never ahead of the physical unit
94% vs 98%, ~30%, week 3, +15%, zero incidents	Locus robots throttled; processing efficiency
6-week → 4-week in 6 weeks, defects ~50%	Agile conversion; QA owned the gate
~\$2.5M, a couple hundred orders	OMS–WMS discrepancy disclosed, not hidden
~\$200K/month per DC, ~\$1M per DC	Two compressed go-lives on one team
\$25 → \$0.40, 98%, 100% citations	LLM triage agent (side build)
11 partners, 10 countries, 48 of 53 in a month	FF App export MVP, April 2022

Leave "\$27M+ cost savings" off the table unless you can show the addition live.

12. Traps

Do not quote an Amazon internal number. You do not know their design rates, ceilings or peak factors. Label every number illustrative and say what replaces it on day one.

Do not say "we" for a decision. "I published the gate. I made the throttle call. I wrote the tolerance."

Do not stack stories. One question, one story, one number. Offer depth; do not volunteer it.

Do not let Situation eat the clock. Fifteen seconds of context. Seventy percent of the story is what you did and what it measured.

Do not open with SAP governance. This room is network and capacity. S/4HANA (S10) is a strong answer to inventory-flow, returns, cutover, and being-wrong questions, and a diluted answer to "tell me about yourself." Use it when they ask about inventory; do not lead with it when they ask about capacity.

Do not invent a headcount. Matrix leadership, said plainly, is fine at L6. A fake org chart is not.

Do not solve the case before clarifying. Sixty seconds of questions out loud is scored, not tolerated.

"What would you do differently" needs a real answer. You have five: shadow one zone first; scheduler heartbeat; measurement audit in week one; dual-path devices; security lead time on the kickoff checklist.

Decide the Bellevue answer before Friday.

13. The morning-of card

1. Read "tell me about yourself" and "why Amazon" (section 3) out loud once. End on the gap sentence.
2. Tell S1 and S2 to a wall in STAR — problem, goal, metric, S, T, A, R, did it move — in under two minutes each. Then say the supply/demand/capacity box for each.
3. Walk Case A on paper: demand → two currencies (cube, hours) → gap by week by building → lever stack with cost → owners and triggers. Re-read the 7A.6 worked example and the 7A.8 U-curve.
4. Do whiteboard math #1, #2, #3 and #6 (7A.11) on paper. Say the five judgment lines in section 7 once.

5. Pick your two questions. Pick your Bellevue sentence.